# Two-loop Type II amplitudes for $gg \to AA$ near threshold and the double parton scattering singularity

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I study the threshold behaviour of the two-loop Type II amplitudes for pseudoscalar pair production, $gg \to AA$, in the heavy top quark limit, in which each pseudoscalar is emitted from its own effective gluon vertex. Their analytic finite remainders are hard to evaluate near threshold because of large cancellations. I expand them in the pseudoscalar velocity $\beta_A$ through $\beta_A^6$ at fixed scattering angle. After the endpoint constants are reduced, all inverse powers of $\beta_A$ and the odd powers through $\beta_A^5$ cancel, and every coefficient is even in $\cos\theta$. At leading power the angle enters only through the logarithm of the transverse momentum $p_T$. I identify the one-loop logarithm with the double parton scattering singularity: with the overall constant fixed at threshold, the factorised formula of Gaunt and Stirling reproduces the velocity dependence of its coefficient in the forward limit to $3 \times 10^{-15}$, and a one-loop calculation of $gg \to HH$ confirms the corresponding weight for a CP-even coupling with the same constant. At two loops the equal and opposite helicity amplitudes grow as $\ln^3 p_T$ and $\ln^2 p_T$; the latter has no logarithm at one loop, and its two-loop logarithm is proportional to the one-loop amplitude. In the finite virtual corrections the one-loop logarithm does not interfere with the tree amplitude, and at two loops the highest logarithm, relative to the squared tree amplitude, is $-\frac{7}{64}\alpha_s^2 N_c^2 \ln^2(2p_T/m_A)$. The released remainders are normalised differently at different orders, with a factor 64 in the two-loop files and the one-loop files given at $\mu_R = 1$ GeV; I convert them to a common normalisation, checked against the renormalisation group equations. For $0 < \beta_A \leq 0.2$ the series reproduce the complete remainders to $6.77 \times 10^{-5}$ at all tested points; they are released with an evaluator.

## I. INTRODUCTION

Precision tests of the Standard Model (SM) and searches for new particles at the Large Hadron Collider (LHC) require theoretical predictions for production rates and differential cross sections. The accuracy of these predictions is important when comparing different models or determining whether a discrepancy with a measurement indicates a new interaction. In gluon fusion processes, QCD radiative corrections can be sizeable, and calculations beyond leading order (LO) are needed to assess their effect on the predicted rates and distributions [1]. Loop amplitudes enter the virtual contributions to these corrections. Their calculation requires not only the evaluation of loop integrals, but also an understanding of their behaviour in the physical region. In particular, amplitudes can develop singularities at special kinematic configurations even when the corresponding tree amplitudes are regular. The location and strength of these singularities matter for numerical evaluation and for the behaviour of differential cross sections [2, 3]. An expansion about such a configuration allows the singular terms to be studied explicitly, together with the regular terms needed for a quantitative prediction.

A relevant example is neutral Higgs boson pair production in two-Higgs-doublet models (2HDM). At the LHC, the dominant production mechanism is gluon fusion through top quark loops, so that loop amplitudes enter already at LO [1, 4]. For pseudoscalar pairs, $gg \to AA$, the next-to-LO (NLO) QCD corrections were first computed in the limit of a heavy top quark [5], later matched to parton showers [6], and subsequently obtained with the full dependence on the top quark mass [1]. In the benchmark considered in Ref. [1], the full NLO cross section for $AA$ lies about 5% below the Born-improved heavy top quark approximation at LHC energies, compared with about 12% for $hH$. For this benchmark, the comparison supports using the heavy top quark limit (HTL) as a starting point for higher orders, with the exact mass dependence retained at LO. Beyond NLO, only the two-loop virtual amplitudes are known in this limit [7]. For SM Higgs pairs, the corrections in the same limit are known to N$^3$LO [8–10]. The same effective gluon interactions also contribute to the production of axion-like particle pairs, and non-resonant $gg \to aa$ production has been used to constrain their interactions at the LHC [11].

In the limit of a heavy top quark, a pseudoscalar field $\Phi^A(x)$ couples to gluons and light quarks through the effective Lagrangian [12]

$$\mathcal{L}_{\text{eff}}^A = \Phi^A(x)\left[-\frac{1}{8}C_G\, O_G(x) - \frac{1}{2}C_J\, O_J(x)\right], \tag{1}$$

with the gluonic and light quark operators

$$O_G(x) = G^{a\mu\nu}\tilde{G}^a_{\mu\nu} = \epsilon_{\mu\nu\rho\sigma}G^{a\mu\nu}G^{a\rho\sigma}, \tag{2}$$

$$G^{a\mu\nu} = \partial^\mu G^{a\nu} - \partial^\nu G^{a\mu} + g_s f^{abc} G^\mu_b G^\nu_c, \tag{3}$$

where $f^{abc}$ are the structure constants of SU(3) and $\epsilon_{\mu\nu\rho\sigma}$ is the Levi-Civita tensor. The light quark operator is the divergence of the flavour singlet axial vector current,

$$O_J(x) = \partial_\mu\left(\bar{\psi}\gamma^\mu\gamma^5\psi\right). \tag{4}$$

By the Adler–Bardeen theorem [13], $C_G$ receives no QCD corrections beyond one loop, whereas $C_J$ starts at second order in the strong coupling $a_s \equiv g_s^2/16\pi^2 = \alpha_s/4\pi$. The Wilson coefficients are

$$C_G(a_s) = -a_s 2^{\frac{5}{4}} G_F^{\frac{1}{2}} \cot\beta, \tag{5}$$

$$C_J(a_s) = -\left[a_s C_F\left(\frac{3}{2} - 3\ln\frac{\mu_R^2}{m_t^2}\right) + a_s^2 C_J^{(2)} + \ldots\right] C_G, \tag{6}$$

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where $G_F$ is the Fermi constant, $m_t$ the top quark mass and $\tan\beta$ the ratio of the vacuum expectation values of the two Higgs doublets, in a model without spontaneous CP violation; this angle is unrelated to the velocity $\beta_A$ of Eq. (8). $C_F$ is the quadratic Casimir of the fundamental representation and $\mu_R$ the scale at which $a_s$ is renormalised. The conventions are those of Ref. [7]. Operator mixing, ultraviolet renormalisation and the treatment of $\gamma_5$ [14] follow the original calculation in Ref. [7]; the same effective theory underlies inclusive pseudoscalar production at NNLO [15].

I consider the dominant gluon fusion subprocess

$$g(p_1) + g(p_2) \to A(p_3) + A(p_4), \tag{7}$$

with incoming gluon momenta $p_{1,2}^2 = 0$ and outgoing pseudoscalar momenta $p_{3,4}^2 = m_A^2$. The amplitude for $gg \to AA$ separates into two classes [7]. In Type I both pseudoscalars originate from a single effective vertex. In Type II each pseudoscalar is emitted from its own vertex and the two vertices are connected by gluon exchange in the $t$ or $u$ channel (Fig. 1a). In the full theory the Type II diagrams correspond to two top quark triangles joined by a gluon, the one-particle reducible contributions of Refs. [1, 5]; for Higgs pairs the corresponding class was computed at two loops in Ref. [9]. The two-loop Type II finite remainders for $gg \to AA$ were obtained in closed form in Ref. [7]. They contain polylogarithms of weight up to four, among them $\mathrm{Li}_{2,2}$, and inverse powers of the Gram determinant $tu - m_A^4$, which vanishes at the pair threshold. The expressions occupy about 204 MB, and large cancellations between their terms make them hard to evaluate close to threshold.

There are two reasons to study this challenging threshold region. The first is phenomenological. In the NLO predictions of Ref. [1] for $m_A = 134.7$ GeV the invariant mass distribution of the pair is larger at $m_{AA} = 300$ GeV, about 30 GeV above threshold, than at 400 GeV, and threshold resummation of this distribution requires at each $m_{AA}$ a hard function obtained from the infrared (IR) finite virtual amplitude [16–20]. The second concerns the analytic structure of loop amplitudes. When the transverse momentum $p_T$ of the final state particles vanishes, a one-loop amplitude with massless internal lines can have a Landau singularity [21] in which each incoming parton splits into two collinear partons that then take part in two separate $2 \to 1$ scatterings. Gaunt and Stirling showed that this double parton scattering (DPS) singularity is at most logarithmic in $p_T$ for SM amplitudes and that, for the processes they considered, it occurs only when the incoming state has $J_z = 0$. They also derived a factorised formula for its coefficient [2]. Dixon and Esterlis studied the corresponding singularity of six-gluon amplitudes in planar $\mathcal{N} = 4$ super Yang–Mills theory to high loop order [3]. As far as I am aware, it has not been examined beyond one loop in QCD at the level of amplitudes for a $2 \to 2$ process with a colour singlet final state; NLO corrections to the underlying splitting mechanism are known for double parton distributions [22]. The Type II amplitude for $gg \to AA$ contains diagrams with this topology: each incoming gluon can split into two gluons, which fuse pairwise into the two pseudoscalars at the effective vertices (Fig. 1b). At threshold, $p_T \to 0$ at every scattering angle, so a threshold expansion probes this configuration directly.

In hadronic collisions, a configuration in which a parton from each proton splits into two can enter both the single and double parton scattering calculations. The overlap must therefore be subtracted to avoid counting the same contribution twice [22, 23]. DPS also receives a contribution in which only one of the two parton pairs arises from a perturbative splitting, while the other proton supplies two partons without such a splitting. This contribution has been studied for Higgs boson pairs [24]. DPS has also been observed in same-sign $W$ pair production at the LHC [25, 26]. The overlap between single and double parton scattering has been re-examined using two-loop splitting kernels [27].

In this paper, I expand the two-loop Type II finite remainders of Ref. [7] in the velocity

$$\beta_A = \sqrt{1 - 4m_A^2/\hat{s}} \tag{8}$$

of the pseudoscalars in the partonic centre of mass frame, at fixed scattering angle, through $\beta_A^6$. The main results are the following.

1. After the constants that arise in the expansion are expressed in a basis of eight transcendental constants, all inverse powers of $\beta_A$ and the odd powers through $\beta_A^5$ cancel, and every coefficient is even in $\cos\theta$, as required by Bose symmetry. At leading power the angle enters only through the logarithm of the transverse momentum $p_T$ of the pseudoscalars.

2. This logarithm is the DPS singularity. At one loop its coefficient follows the factorised formula of Ref. [2] for gluon splitting and the opposite signs of the amplitudes for $g^+g^+ \to A$ and $g^-g^- \to A$: with the overall constant fixed at threshold, it reproduces the velocity dependence of the exact one-loop coefficient in the forward limit. A one-loop calculation of $gg \to HH$ confirms the corresponding weight for a CP-even coupling.

3. At two loops the $J_z = 0$ amplitude, with equal gluon helicities, grows as $\ln^3 p_T$, and the ratio of the leading logarithms at two and one loops is the one found to all orders in planar $\mathcal{N} = 4$ super Yang–Mills theory [3]. The $|J_z| = 2$ amplitude, with opposite helicities, has no DPS logarithm at one loop but grows as $\ln^2 p_T$ at two loops; in the forward limit this logarithm is the one-loop amplitude multiplied by $-4N_c a_s \ln^2(2p_T/m_A)$ at every velocity tested, and its origin remains open.

4. In the finite virtual corrections to the cross section the one-loop logarithm cancels. At two loops $\ln^3 p_T$ cancels, and the highest power is $\ln^2 p_T$.

All results are given in a consistent normalisation. The series reproduce the unexpanded remainders with relative errors of at most $6.77 \times 10^{-5}$ for $\beta_A \leq 0.2$; they are released together with an evaluator [28].

The paper is organised as follows. Section II sets out the conventions, the normalisation of the remainders and the

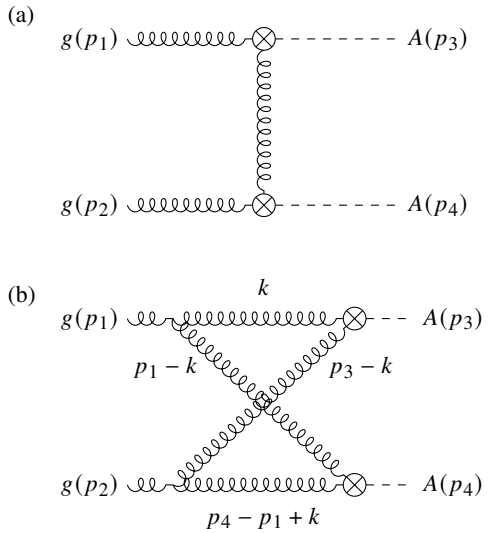



FIG. 1. (a) Type II tree diagram in the $t$ channel; the $u$-channel diagram is obtained by exchanging the pseudoscalars. The crossed circles are the effective vertices of $O_G$. (b) One-loop Type II diagram with loop momentum $k$. Its double parton scattering singularity lies at $k = xp_1$ and $p_3 - k = (1-x)p_2$, where the four internal gluons are on shell and collinear with the incoming gluons: each incoming gluon splits into two, which fuse pairwise at the effective vertices. Gaunt and Stirling call this topology the crossed box [2].

counting of orders. Section III summarises the procedure and describes the expansion and its accuracy. Section IV discusses the leading power terms and their relation to the DPS singularity, and Sec. V their consequences for the cross section. The conclusions are in Sec. VI.

## II. AMPLITUDES AND CONVENTIONS

### A. Theoretical framework

To study the threshold behaviour of $gg \to AA$ through two loops, I use the amplitudes calculated in Ref. [7] with the effective Lagrangian of Eq. (1). The labels Type I and Type II follow that reference and distinguish the two classes of diagrams. They are unrelated to the Yukawa types of two-Higgs-doublet models. The Type II tree diagrams correspond, in the full theory, to two top quark triangles connected by a gluon, i.e. the one-particle reducible contribution of Ref. [1]. Here I consider the contribution with two $O_G$ insertions and omit the superscript II from the amplitudes. Diagrams with one $O_G$ and one $O_J$ insertion first contribute at one loop. Their coefficients are proportional to the $O_G$–$O_G$ tree coefficients, as discussed in Sec. II D.

### B. Kinematics

The Mandelstam variables are

$$\hat{s} = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_2 - p_3)^2, \quad (9)$$

with $\hat{s} + t + u = 2m_A^2$. Following the conventions of Ref. [7], I use

$$\hat{s} = m_A^2 \frac{(1+x)^2}{x}, \qquad t = -m_A^2 y, \qquad u = -m_A^2 z, \qquad (10)$$

with $x + x^{-1} = y + z$. With $c = \cos\theta$ in the partonic centre of mass frame,

$$\hat{s} = \frac{4m_A^2}{1-\beta_A^2}, \qquad t, u = m_A^2 - \frac{\hat{s}}{2}(1 \mp \beta_A c), \qquad (11)$$

and the threshold substitution is

$$x = \frac{1-\beta_A}{1+\beta_A}, \qquad y, z = \frac{1+\beta_A^2 \mp 2\beta_A c}{1-\beta_A^2}. \qquad (12)$$

All three variables tend to one as $\beta_A \to 0^+$ at fixed $c$, and the expansion has to follow this correlated limit. The Gram determinant and the transverse momentum are

$$tu - m_A^4 = \frac{4m_A^4\beta_A^2(1-c^2)}{(1-\beta_A^2)^2}, \qquad p_T^2 = \frac{m_A^2\beta_A^2(1-c^2)}{1-\beta_A^2}. \quad (13)$$

Both vanish at threshold at every angle, and also at fixed $\beta_A$ when $c \to \pm 1$.

### C. Tensor decomposition and helicity amplitudes

With the overall factor $C_G^2$ removed, the amplitude is decomposed as

$$\mathcal{M}_{ab}^{\mu\nu} = \delta_{ab}\left(T_1^{\mu\nu}\mathcal{M}_1 + T_2^{\mu\nu}\mathcal{M}_2\right), \qquad (14)$$

with

$$T_1^{\mu\nu} = g^{\mu\nu} - \frac{p_1^\nu p_2^\mu}{p_1\cdot p_2}, \qquad (15)$$

$$T_2^{\mu\nu} = g^{\mu\nu} + \frac{1}{(p_1\cdot p_2)p_T^2}\Big[m_A^2 p_2^\mu p_1^\nu - 2(p_1\cdot p_3)p_2^\mu p_3^\nu - 2(p_2\cdot p_3)p_3^\mu p_1^\nu + 2(p_1\cdot p_2)p_3^\mu p_3^\nu\Big], \qquad (16)$$

$$p_T^2 = \frac{tu - m_A^4}{\hat{s}}. \qquad (17)$$

For physical gluon polarisations, $T_1$ contributes only to equal helicities ($J_z = 0$) and $T_2$ only to opposite helicities ($|J_z| = 2$), each with a contraction of unit modulus. Up to phases the helicity amplitudes are [29]

$$\mathcal{A}_{\pm\pm} = C_G^2\,\mathcal{M}_1, \qquad \mathcal{A}_{\pm\mp} = C_G^2\,\mathcal{M}_2\,e^{\pm 2i\phi}, \qquad (18)$$

where $\phi$ is the azimuth of $\vec{p}_3$, and $\sum_{\text{hel,col}}|\mathcal{A}|^2 = 2(N_c^2-1)\,C_G^4\,(|\mathcal{M}_1|^2 + |\mathcal{M}_2|^2)$. The labels 1 and 2 refer to the tensors, not to the diagram classes.

At tree level the two Type II diagrams give

$$\mathcal{M}_1^{(0)} = \frac{i\hat{s}}{2},$$

$$\mathcal{M}_2^{(0)} = \frac{im_A^2(x-z)(x^2+1)(xz-1)}{4xz(x^2-xz+1)} = 2im_A^2\beta_A^2(1-c^2) + O(\beta_A^4), \qquad (19)$$

in the variables of Eq. (10). I obtained Eq. (19) from the projections of the original calculation and checked it against a direct evaluation of the two Feynman diagrams with the vertex $iC_G\varepsilon_{\mu\alpha\rho\beta}k_1^\mu k_2^\rho\delta^{ab}$ that follows from Eq. (1) (App. B). The suppression of $\mathcal{M}_2^{(0)}$ by $\beta_A^2$ is the centrifugal barrier of the $|J_z| = 2$ state.

### D. Finite remainders and normalisation

The coefficients $\mathcal{M}_i$ of Eq. (18), from which $C_G^2$ has been factored out, are expanded as

$$\mathcal{M}_i = \mathcal{M}_i^{(0)} + a_s\mathcal{M}_i^{(1)} + a_s^2\mathcal{M}_i^{(2)} + \cdots, \tag{20}$$

with $a_s$ renormalised at $\mu_R$. The finite remainders, called the finite part in Ref. [7], are defined by the IR subtraction of Refs. [7, 30],

$$\begin{aligned}\mathcal{M}^{(1)} &= 2I^{(1)}\mathcal{M}^{(0)} + \mathcal{M}^{(1),\text{fin}},\\ \mathcal{M}^{(2)} &= 4I^{(2)}\mathcal{M}^{(0)} + 2I^{(1)}\mathcal{M}^{(1)} + \mathcal{M}^{(2),\text{fin}}.\end{aligned} \tag{21}$$

From here on $\mathcal{M}^{(n)}$ denotes the finite remainder $\mathcal{M}^{(n),\text{fin}}$.

At two loops, I decompose each finite remainder into powers of the number of light quark flavours,

$$\mathcal{M}_i^{(2)} = \mathcal{M}_{i,0}^{(2)} + n_f\,\mathcal{M}_{i,1}^{(2)} + n_f^2\,\mathcal{M}_{i,2}^{(2)}. \tag{22}$$

Each $\mathcal{M}_{i,k}^{(2)}$ is finite, but none is the amplitude of a separate process: the term linear in $n_f$, for example, combines the diagrams with a closed quark loop with the terms of the renormalisation and of the infrared subtraction that depend on $n_f$. In the rest of the paper, the complete coefficient means $\mathcal{M}_i^{(2)}$ at $n_f = 5$. Furthermore, the remainders released with Ref. [7] are normalised differently at different orders: the two-loop remainders carry an overall factor 64 relative to the tree and one-loop amplitudes, and the one-loop remainders are those for $\mu_R = 1$ GeV. In the normalisation of Eq. (20) the remainders are therefore

$$\begin{aligned}\mathcal{M}_i^{(1)} &= \mathcal{M}_{i,\text{publ}}^{(1)} + 2\beta_0 \ln\frac{\mu_R^2}{\text{GeV}^2}\,\mathcal{M}_i^{(0)},\\ \mathcal{M}_i^{(2)} &= \frac{1}{64}\,\mathcal{M}_{i,\text{publ}}^{(2)},\end{aligned} \tag{23}$$

with $\beta_0 = (11N_c - 2n_f)/3$ and masses in GeV in the published expressions. The one-loop term adds the scale dependence required by renormalisation group invariance; with it, the released one and two-loop remainders are consistent with each other at every tested point. Appendix B gives the evidence. The logarithmic structure studied below, the expansion and its accuracy do not depend on this conversion; the one-loop constant and the finite virtual corrections of Sec. V do.

The one-loop amplitude with one $O_G$ and one $O_J$ insertion enters at $O(a_s^4)$, the order of the two-loop $O_G$–$O_G$ amplitude, since $C_J = O(a_s)\,C_G$. It is proportional to the tree amplitude: in the normalisation of Eqs. (3.6) and (3.7) of Ref. [7],

$$\mathcal{M}_{i,GJ,\text{publ}}^{(1),\text{fin}} = -48\,n_f\,\mathcal{M}_i^{(0)}, \qquad i = 1, 2, \tag{24}$$

at every point of the physical phase space. The proportionality is consistent with the axial anomaly, which relates the divergence of the flavour singlet axial current to the gluonic pseudoscalar operator [13, 14, 31]. At $O(a_s^4)$ the mixed contribution therefore has the kinematic dependence of the tree amplitude and adds no logarithm of $p_T$.

### E. Perturbative order

The expansion in Eq. (20) is written for the coefficients $\mathcal{M}_i$, with the overall factor $C_G^2$ removed. The superscript $(n)$ counts loops of gluons and light quarks in the effective theory; in a two-Higgs-doublet model the top quark loops are contained in the Wilson coefficients. The perturbative order of a contribution to the cross section also depends on the powers of $a_s$ in the Wilson coefficients.

Since $C_G \propto a_s$, the Type II tree, one-loop and two-loop contributions to the full amplitude are of order $a_s^2$, $a_s^3$ and $a_s^4$, respectively. In a 2HDM, the leading amplitude is of Type I and starts at $O(a_s)$, giving a LO cross section of order $a_s^2$. Interference between the leading Type I amplitude and the Type II tree amplitude contributes at order $a_s^3$, that is at NLO. This is the one-particle reducible contribution to the virtual corrections in Refs. [1, 5]. The Type II two-loop amplitude first enters through its interference with the leading Type I amplitude at order $a_s^5$, or N$^3$LO. The same counting applies to the corresponding class of amplitudes for Higgs pairs [9].

The mixed $O_G$–$O_J$ contribution discussed in Sec. II D also contributes at this N$^3$LO. In the heavy top quark theory, $C_J = O(a_s)\,C_G$ [7], so its one-loop amplitude is of order $a_s^4$, the same as the two-loop $O_G$–$O_G$ amplitude.

When no Type I amplitude is present, the Type II tree amplitude instead gives the LO cross section. This is the case for an axion-like particle whose only relevant interaction is its coupling to gluons through $G\tilde{G}$. The one-loop and two-loop terms in Eq. (20) then provide virtual contributions at NLO and NNLO, respectively. The present results therefore supply the $O_G$–$O_G$ part of the two-loop virtual correction at NNLO for this process. A contact interaction of dimension six containing two pseudoscalars and two gluon fields would introduce an additional Type I amplitude, as in Ref. [11].

### F. Finite top quark mass effects

The threshold expansion and the heavy top quark approximation involve different limits. Taking $\beta_A \to 0$ places the pair invariant mass close to $2m_A$, but imposes no restriction on $m_A/m_t$. Finite top quark mass effects must therefore be assessed separately when $A$ is the pseudoscalar of a 2HDM, at values of $\tan\beta$ for which the top quark loop dominates. No heavy top quark approximation is involved when the coupling to $G\tilde{G}$ is specified directly as an effective interaction, as in the axion-like particle case above.

For the Type II tree amplitude, the dependence on the top quark mass is known. Each triangle in the full theory couples a pseudoscalar to one on-shell gluon and one virtual gluon. The

two triangles in a given channel have the same form factor, so the contribution of each channel is multiplied, relative to its heavy top quark limit, by

$$\left[\frac{I_2(\tau_A,\lambda)}{I_2^\infty}\right]^2, \quad \tau_A=\frac{4m_t^2}{m_A^2}, \quad \lambda=\frac{4m_t^2}{t} \quad \text{or} \quad \frac{4m_t^2}{u}. \tag{25}$$

Here $I_2$ is the loop function that also appears in $A\to Z\gamma$, given in Ref. [1], and $I_2^\infty$ denotes its heavy top quark limit. At the pair threshold, $t=u=-m_A^2$, so both channels have the same factor: 1.006, 1.013 and 1.022 for $m_A=125$, 150 and 170 GeV, respectively, with $m_t=172.5$ GeV. These differ from unity by at most about 2.2%.

For the loop corrections, a partial estimate can be made for the one-loop DPS logarithm. Its coefficient involves the amplitudes for $gg\to A$ with both gluons on shell, as discussed in Sec. IV C. The relevant triangle form factor is therefore obtained in the limit $\lambda\to\infty$, rather than at the spacelike virtuality of the exchanged gluon in the tree diagrams:

$$\frac{I_2(\tau_A,\infty)}{I_2^\infty}=\tau_A f(\tau_A), \qquad f(\tau)=\arcsin^2\!\left(\frac{1}{\sqrt{\tau}}\right). \tag{26}$$

If the DPS factorisation applies to the full theory, with the top quark triangles treated as hard subgraphs, the coefficient of the one-loop logarithm is multiplied by $[\tau_A f(\tau_A)]^2$. This gives 1.096, 1.145 and 1.196 for the same three masses, while the velocity dependence and the helicity selection rule would remain unchanged. Neither this estimate nor the comparison of tree amplitudes determines the finite top quark mass corrections to the remaining one-loop terms or to the complete two-loop amplitude.

## III. THRESHOLD EXPANSION

The calculation has four steps, each checked independently. First, the finite remainders of Ref. [7] are split into their flavour coefficients and brought to the normalisation of Eq. (23) (App. B). Second, each flavour coefficient is expanded about $\beta_A=0$ at fixed angle, through $\beta_A^4$ and, independently, through $\beta_A^6$ (Sec. III A). Third, the constants that arise at the endpoints are reduced to a basis of eight transcendental constants, after which the inverse and odd powers of $\beta_A$ cancel identically (Sec. III B and App. A). Fourth, the series are compared with the unexpanded remainders, evaluated in Mathematica at increasing working precision and, at selected points, with an independent Python code (Sec. III C). The leading power terms obtained in this way are analysed in Sec. IV, where the forward limit of the one-loop remainders and a one-loop calculation of $gg\to HH$ test their relation to the DPS singularity, and their consequences for the cross section in Sec. V. The software and the computing environment are described in App. D.

### A. Construction

The remainders of Ref. [7] are functions of $x$, $y$ and $z$, Eq. (10). I replace these variables by their expressions in $\beta_A$ and $c$ from Eq. (12), and expand each flavour coefficient about $\beta_A=0$ at fixed $c$,

$$\mathcal{M}^{(2)}_{i,r}(\beta_A,c)=\sum_{n=-4}^{p}\sum_k C^{(i,r)}_{nk}\,\beta_A^n\ln^k\beta_A+O_{\log}(\beta_A^{p+1}), \tag{27}$$

with $r=0,1,2$ and $p=4$ or 6. The coefficients $C^{(i,r)}_{nk}$ depend on $c$, $m_A^2$, $\mu_R^2$ and $N_c$, and $O_{\log}(\beta_A^n)$ stands for $\beta_A^n$ times a polynomial in $\ln\beta_A$. Inverse powers appear in intermediate steps because the tensor basis and the rational prefactors contain inverse Gram determinants, Eq. (13). They must be kept until all contributions at a given power have been combined.

The remainders contain logarithms, classical polylogarithms and the function $\mathrm{Li}_{2,2}$ of weight four [32, 33], whose arguments approach correlated endpoints as $\beta_A\to 0$. I expand the rational prefactors exactly and obtain the expansions of the special functions from differential equations in $\beta_A$, with boundary values at $\beta_A=0$. These equations have the iterated-integral form familiar from harmonic polylogarithms [34] and from canonical differential equations for Feynman integrals [35]; the map $x=(1-\beta_A)/(1+\beta_A)$ is one of the transformations under which harmonic polylogarithms are closed [34]. The method thus expands the known remainders, not the loop integrals, which could instead be expanded near threshold by the method of regions [36]. For the dilogarithm, for example,

$$\mathrm{d}\,\mathrm{Li}_2(x)/\mathrm{d}\beta_A=2(1-\beta_A^2)^{-1}\ln[2\beta_A/(1+\beta_A)] \text{ with} \tag{28}$$

$$\mathrm{Li}_2(1)=\pi^2/6 \text{ gives} \tag{29}$$

$$\mathrm{Li}_2(x)=\frac{\pi^2}{6}+2\beta_A\big[\ln(2\beta_A)-1\big]-\beta_A^2 +\beta_A^3\left[\frac{2}{3}\ln(2\beta_A)+\frac{1}{9}\right]-\frac{2}{3}\beta_A^4+O\Big(\beta_A^5\ln\beta_A\Big). \tag{30}$$

Since the prefactors start at $\beta_A^{-4}$, the special functions have to be expanded four orders beyond the final accuracy. The fourth and sixth order expansions were constructed independently, and all coefficients common to both agree.

### B. Endpoint constants and exact cancellations

The boundary values at $\beta_A=0$ introduce thirteen constants $\mathrm{MB}[p,q,a,w]$, defined in Eq. (A1) of App. A. Their arguments take the values $\pm1$ and $\frac{1}{2}$, so they are alternating Euler sums of weight three and four [37]. An integer relation search with the PSLQ algorithm [38] at 110 digits expresses each of them in the basis

$$\{\zeta_3,\ \pi^2\ln 2,\ \ln^3 2,\ \pi^4,\ \pi^2\ln^2 2,\ \ln^4 2,\ \mathrm{Li}_4(\tfrac{1}{2}),\ \zeta_3\ln 2\}, \tag{31}$$

with a largest residual of $4\times10^{-107}$ (Table VII).

With these reductions, and with the dilogarithms of $(1-c)/2$, $(c-1)/(c+1)$ and $(c+1)/(c-1)$ rewritten through $\mathrm{Li}_2\big((1+c)/2\big)$ by the reflection and Landen identities, every

coefficient $C_{nk}^{(i,r)}$ becomes a polynomial in

$$\left\{\ln\beta_A,\ \ln(1\pm c),\ \mathrm{Li}_2\left(\tfrac{1+c}{2}\right),\ \ln\tfrac{m_A^2}{\mu_R^2}\right\}\cup\{\ln 2,\ \pi,\ \zeta_3,\ \mathrm{Li}_4(\tfrac{1}{2})\} \tag{32}$$

with coefficients rational in $c$ and $N_c$. Treating these quantities as algebraically independent, I find that

1. the coefficients of $\beta_A^{-4},\ldots,\beta_A^{-1}$ vanish identically in every flavour coefficient;

2. the coefficients of $\beta_A$, $\beta_A^3$ and $\beta_A^5$ vanish identically, so that through $\beta_A^6$ the expansion is a series in $\beta_A^2$ with logarithms;

3. the coefficients of $\beta_A^0$, $\beta_A^2$, $\beta_A^4$ and $\beta_A^6$ of all six flavour coefficients, 24 in total, are even in $c$, so that Bose symmetry holds order by order.

The cancellations and the reductions of the constants were confirmed independently in Mathematica.

The cancellations have a simple origin. Exchanging the pseudoscalars exchanges $t$ and $u$, and the invariants that are symmetric under this exchange,

$$\hat{s}=\frac{4m_A^2}{1-\beta_A^2},\qquad (t-u)^2=\hat{s}^2\beta_A^2c^2,\qquad tu-m_A^4=\hat{s}\,p_T^2, \tag{33}$$

with $p_T^2=\hat{s}\beta_A^2(1-c^2)/4$, depend on $\beta_A$ only through $\beta_A^2$. The loops contain no massive particles, so there is no threshold singularity at $\hat{s}=4m_A^2$, and the colourless final state has no Coulomb singularity. Near threshold, the only singularity is then the one at $p_T\to 0$ discussed in Sec. IV C, which is logarithmic at one loop [2]. The Gram determinant $tu-m_A^4$ vanishes at the same point, but its inverse powers come from the tensor basis and the rational prefactors and are not singularities of the amplitude. A remainder of the form $\sum_k a_k \ln^k p_T^2$, with coefficients $a_k$ that are analytic in the invariants of Eq. (33) at threshold, is a series in $\beta_A^2$ with logarithms and has no inverse powers. This explains the form of the cancellations found above, but at two loops it is not an independent proof, since there the logarithmic nature of the singularity is established only by the calculation itself. Table I lists the leading terms. Since the odd powers vanish, the first omitted term is expected to be $O_{\log}(\beta_A^{p+2})$ rather than the $O_{\log}(\beta_A^{p+1})$ of Eq. (27), so the truncation errors should scale as $\beta_A^6$ and $\beta_A^8$ for $p=4$ and $p=6$, respectively, up to logarithmic factors.

The coefficient of $n_f^2$ is the simplest:

$$\mathcal{M}_{1,2}^{(2)}(\beta_A,c)=K+O(\beta_A^2), \tag{34}$$

$$\mathcal{M}_{2,2}^{(2)}(\beta_A,c)=\beta_A^2(1-c^2)K+O(\beta_A^4), \tag{35}$$

where $K$ does not depend on the angle. It is the only flavour coefficient that follows the tree-level pattern $\mathcal{M}_2\propto\beta_A^2(1-c^2)$ of Eq. (19), with the same ratio to the tree amplitude in both tensor structures. The complete expressions through $\beta_A^6$ are released as Wolfram Language files (App. D).

TABLE I. Leading terms of the threshold expansion of the two-loop Type II remainders. The second column lists the inverse powers with nonzero coefficients in the construction files; they cancel identically after the reductions of Sec. III B, as do all odd powers. The last column gives the powers of $\mathcal{L}$, Eq. (37), at leading power.

| quantity | inverse powers in construction | leading power | powers of $\mathcal{L}$ |
|---|---|---|---|
| $\mathcal{M}_{1,0}^{(2)}$ | $\beta_A^{-2},\beta_A^{-1}$ | $\beta_A^0$ | 0, 1, 2, 3 |
| $\mathcal{M}_{1,1}^{(2)}$ | – | $\beta_A^0$ | 0, 1, 2 |
| $\mathcal{M}_{1,2}^{(2)}$ | – | $\beta_A^0$ | 0 |
| $\mathcal{M}_{2,0}^{(2)}$ | $\beta_A^{-4},\beta_A^{-3},\beta_A^{-2},\beta_A^{-1}$ | $\beta_A^0$ | 0, 1, 2 |
| $\mathcal{M}_{2,1}^{(2)}$ | $\beta_A^{-1}$ | $\beta_A^0$ | 0, 1 |
| $\mathcal{M}_{2,2}^{(2)}$ | – | $\beta_A^2$ | 0 |

### C. Accuracy

I compared the series with the unexpanded remainders on the grid in Table II, which contains 9 values of $\beta_A$ between $10^{-3}$ and 3/10, 5 scattering angles and 3 combinations of mass, renormalisation scale and number of colours. At each of the 135 points, the comparison included the six flavour coefficients and the two complete coefficients, giving 1080 comparisons in total. Each unexpanded remainder was evaluated with 80, 120, 160, 220 and 300 digits of working precision, and a value was accepted once two successive precisions agreed to a relative difference below $10^{-25}$, with at least 30 significant digits in both. The parameter set with $N_c\neq 3$ was included to test the colour dependence; all physical results use $N_c=3$. The relative truncation error is defined as

$$E_i^{(p)}=\frac{\left|\mathcal{M}_i^{[p]}-\mathcal{M}_i\right|}{|\mathcal{M}_i|},\qquad p=4,6, \tag{36}$$

where $\mathcal{M}_i$ is the unexpanded remainder and $\mathcal{M}_i^{[p]}$ its series through $\beta_A^p$. Figure 2 shows these errors for the complete coefficients at $n_f=5$, $\mathcal{M}_1$ in the left column and $\mathcal{M}_2$ in the right. At each $\beta_A$ the maximum over the angles and parameter sets of Table II is plotted, with circles for the series through $\beta_A^4$ and squares for the series through $\beta_A^6$. For the series through $\beta_A^6$ the maximum comes from the same point at every velocity and for both coefficients, $c=0$ with $m_A=150$ GeV, $\mu_R=75$ GeV and $N_c=4$. These curves therefore follow the error of a single point as $\beta_A$ varies, and the slopes and scaled errors discussed below are not affected by a change of the point that gives the maximum. At each velocity and for both orders the errors at the other points are smaller by at most a factor of eight, so the maximum is not set by an isolated point.

For a series in $\beta_A^2$, the first omitted terms are of order $\beta_A^6$ and $\beta_A^8$, up to powers of $\ln\beta_A$; the reference lines in the upper panels show these powers. Straight-line fits of the logarithm of the maximum error against $\ln\beta_A$ over the plotted range give slopes of 6.005 and 8.005 for $\mathcal{M}_1$, and 5.955 and 8.034 for $\mathcal{M}_2$, for the series through $\beta_A^4$ and $\beta_A^6$.

The lower panels divide the maximum errors by the expected

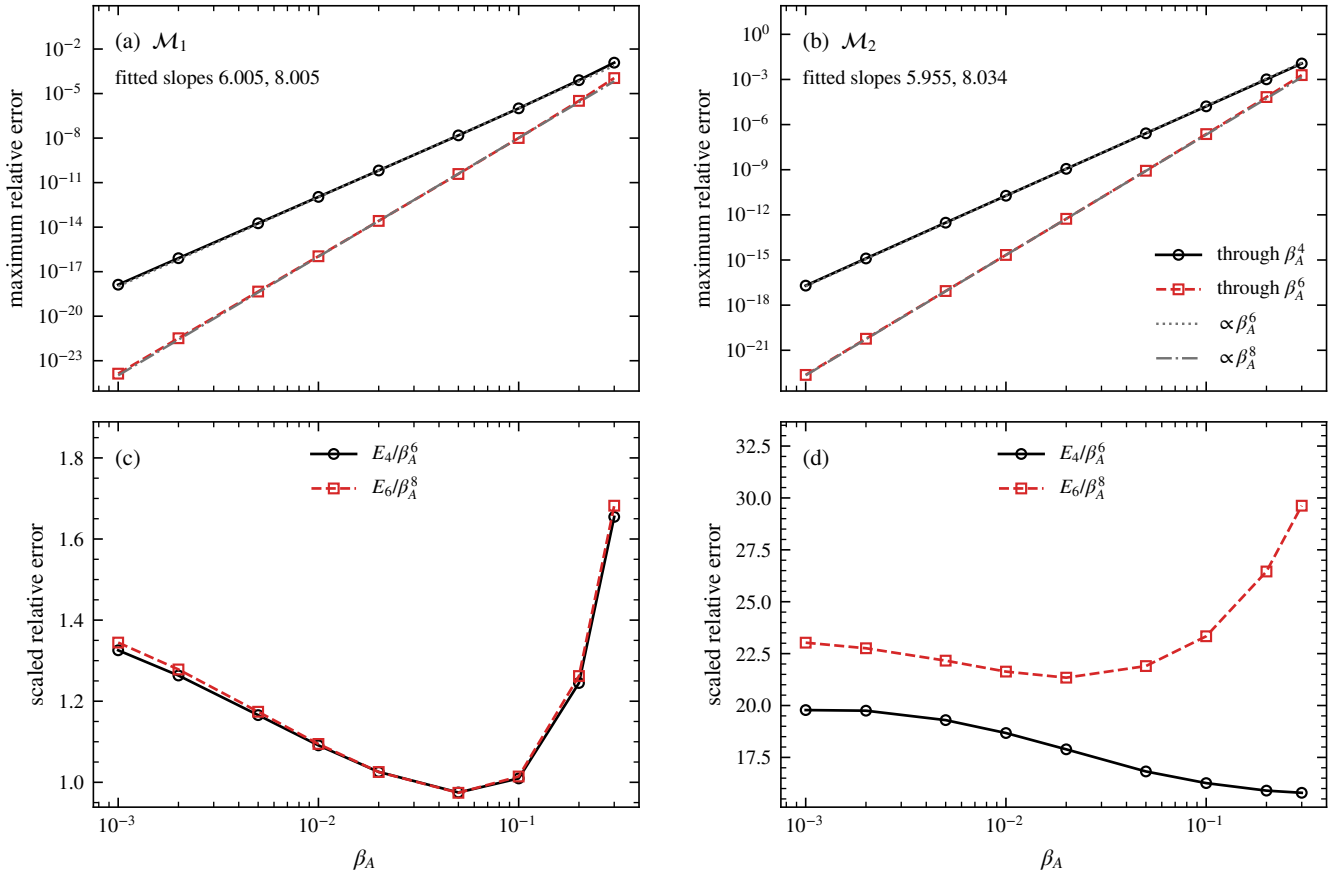



FIG. 2. Relative truncation errors, Eq. (36), of the complete two-loop coefficients at $n_f = 5$: $\mathcal{M}_1$ (left) and $\mathcal{M}_2$ (right). Upper panels: maximum over the angles and parameter sets of Table II for the series through $\beta_A^4$ (circles) and through $\beta_A^6$ (squares), with lines proportional to $\beta_A^6$ and $\beta_A^8$. Lower panels: the same maxima divided by $\beta_A^6$ and $\beta_A^8$.

powers, $\beta_A^6$ for the series through $\beta_A^4$ and $\beta_A^8$ for the series through $\beta_A^6$, so that exact power behaviour would give horizontal lines. The ratios vary slowly, because both the omitted terms and the remainders that normalise the errors contain powers of $\ln \beta_A$. An incorrect coefficient at a retained order would instead make them grow as an inverse power of $\beta_A$, which is not seen anywhere in the tested range. Towards $\beta_A = 0.3$, the scaled errors of the series through $\beta_A^6$ rise for both coefficients.

Table III lists the maximum errors at selected velocities. For $\beta_A \le 0.2$ the series through $\beta_A^6$ reproduces $\mathcal{M}_1$ and $\mathcal{M}_2$ with relative errors of at most $3.23 \times 10^{-6}$ and $6.77 \times 10^{-5}$; at $\beta_A = 0.3$ the errors are $1.10 \times 10^{-4}$ and $1.94 \times 10^{-3}$. Bose symmetry was checked by comparing the values at $c$ and $-c$: they are identical in all stored digits in 432 comparisons of the unexpanded coefficients and 864 of the series.

The reference values were computed in Mathematica from the published expressions. To check their evaluation independently, I wrote a second code in Python with mpmath [39]. It reads the published files with its own parser and computes $\mathrm{Li}_{2,2}$ from its defining double sum, or from a one-dimensional integral representation where the sum converges slowly. At four points covering the three parameter sets, the two complete coefficients agree with the reference values to a relative difference of at most $1.3 \times 10^{-40}$, the precision to which the reference values are stored. The six flavour coefficients, compared at $\beta_A = 1/10$ and $c = 1/3$, agree to $1.6 \times 10^{-40}$. At $c = 0$, where $t = u$, the published expressions have removable singularities and are evaluated as limits.

TABLE II. Validation grid. At each point the six flavour coefficients and the two complete $n_f = 5$ coefficients are compared.

| quantity | values |
|---|---|
| $\beta_A$ | 1/1000, 1/500, 1/200, 1/100, 1/50, 1/20, 1/10, 1/5, 3/10 |
| $c$ | $-2/3, -1/3, 0, 1/3, 2/3$ |
| $(m_A/\mathrm{GeV}, \mu_R/\mathrm{GeV}, N_c)$ | (125, 125, 3), (125, 250, 3), (150, 75, 4) |

The expansion is made at fixed angle, but $p_T$ also vanishes as $c \to \pm 1$ at fixed $\beta_A$, so the series has to be tested close to the beam as well. I therefore compared it with the unexpanded remainders at $c = 0.9, 0.99, 0.999$ and $-0.99$ for $\beta_A = 0.1, 0.2$ and 0.3. Table IV gives the relative errors of the series through $\beta_A^6$. All values are below the maximum on the main grid at

TABLE III. Maximum relative errors of the complete Type II coefficients on the validation grid. At each value of $\beta_A$, the maximum is taken over the 5 angles and 3 parameter sets of Table II.

| | $\mathcal{M}_1$ through | | $\mathcal{M}_2$ through | |
|---|---|---|---|---|
| $\beta_A$ | $\beta_A^4$ | $\beta_A^6$ | $\beta_A^4$ | $\beta_A^6$ |
| 0.01 | $1.09\times10^{-12}$ | $1.09\times10^{-16}$ | $1.87\times10^{-11}$ | $2.16\times10^{-15}$ |
| 0.05 | $1.52\times10^{-8}$ | $3.81\times10^{-11}$ | $2.63\times10^{-7}$ | $8.55\times10^{-10}$ |
| 0.10 | $1.01\times10^{-6}$ | $1.01\times10^{-8}$ | $1.63\times10^{-5}$ | $2.33\times10^{-7}$ |
| 0.20 | $7.97\times10^{-5}$ | $3.23\times10^{-6}$ | $1.02\times10^{-3}$ | $6.77\times10^{-5}$ |
| 0.30 | $1.21\times10^{-3}$ | $1.10\times10^{-4}$ | $1.15\times10^{-2}$ | $1.94\times10^{-3}$ |

TABLE IV. Relative truncation errors $E_i^{(6)}$ of the complete $n_f = 5$ coefficients towards the beam, at $m_A = \mu_R = 125\,\text{GeV}$ and $N_c = 3$. The last rows give the largest value on the grid of Table II. The values at $c = -0.99$ equal those at $-c$.

| | $c$ | $\beta_A = 0.1$ | $\beta_A = 0.2$ | $\beta_A = 0.3$ |
|---|---|---|---|---|
| $\mathcal{M}_1$ | 0.9 | $6.36\times10^{-9}$ | $1.11\times10^{-6}$ | $1.78\times10^{-5}$ |
| $\mathcal{M}_1$ | 0.99 | $1.06\times10^{-8}$ | $2.43\times10^{-6}$ | $5.76\times10^{-5}$ |
| $\mathcal{M}_1$ | 0.999 | $1.23\times10^{-8}$ | $2.96\times10^{-6}$ | $7.18\times10^{-5}$ |
| $\mathcal{M}_2$ | 0.9 | $1.39\times10^{-7}$ | $3.08\times10^{-5}$ | $6.68\times10^{-4}$ |
| $\mathcal{M}_2$ | 0.99 | $6.68\times10^{-8}$ | $1.47\times10^{-5}$ | $2.88\times10^{-4}$ |
| $\mathcal{M}_2$ | 0.999 | $6.21\times10^{-8}$ | $1.65\times10^{-5}$ | $3.74\times10^{-4}$ |
| $\mathcal{M}_1$ | grid max. | $1.01\times10^{-8}$ | $3.23\times10^{-6}$ | $1.10\times10^{-4}$ |
| $\mathcal{M}_2$ | grid max. | $2.33\times10^{-7}$ | $6.77\times10^{-5}$ | $1.94\times10^{-3}$ |

the same $\beta_A$, except for $\mathcal{M}_1$ at $\beta_A = 0.1$, where the largest value, $1.23\times10^{-8}$ at $c = 0.999$, exceeds it by 21%. The series thus remains accurate close to the beam, consistent with its coefficients having no poles at $c = \pm1$, only logarithms of $1 \pm c$ (Sec. IV A).

## IV. LEADING POWER AND THE DOUBLE PARTON SCATTERING SINGULARITY

### A. Leading power

To determine the leading threshold behaviour of the two-loop remainders, I first collect the terms of order $\beta_A^0$, including all powers of $\ln\beta_A$ that multiply them. The notation $\big|_{\beta_A^0}$ used below denotes this part of the expansion, rather than the value of the remainder at $\beta_A = 0$.

In the explicit result, the logarithms of $\beta_A$ and the angular logarithms combine through $\ln\beta_A + \frac{1}{2}\ln(1-c^2)$. It is convenient to introduce

$$\begin{aligned}\mathcal{L} &= \ln\beta_A + \tfrac{1}{2}\ln(1-c^2) + \ln 2 + \frac{i\pi}{2}\\ &= \tfrac{1}{2}\ln\bigl[-4\beta_A^2(1-c^2)\bigr],\end{aligned} \tag{37}$$

where the second equality uses the branch $\ln(-X) = \ln X + i\pi$ for $X > 0$. As shown below, $\mathcal{L}$ is the logarithm of the transverse momentum of the pseudoscalars, $\ln(2p_T/m_A)$, up to the constant phase $i\pi/2$ and terms of order $\beta_A^2$. The relation of $\mathcal{L}$ to the physical kinematics follows from Eq. (13). Substituting the expressions for $tu - m_A^4$ and $p_T^2$ gives

$$\begin{aligned}\mathcal{L} &= \tfrac{1}{2}\ln\frac{m_A^4 - tu}{m_A^4} + \ln(1-\beta_A^2)\\ &= \ln\frac{2p_T}{m_A} + \frac{i\pi}{2} + \tfrac{1}{2}\ln(1-\beta_A^2).\end{aligned}$$

Since $\ln(1-\beta_A^2) = O(\beta_A^2)$, the last term contributes only beyond leading power. Thus, near threshold, the combination of velocity and angular logarithms in Eq. (37) is the logarithm of the transverse momentum, up to the constant phase $i\pi/2$ and power corrections.

When the calculated leading power terms are rewritten in terms of $\mathcal{L}$, no separate $\ln(1\pm c)$ or $\mathrm{Li}_2\bigl((1+c)/2\bigr)$ remains. The rational dependence on $c$ also cancels. The angular dependence is therefore entirely contained in $\mathcal{L}$, while the coefficients of its powers are independent of the scattering angle. For general $N_c$ and $n_f$, the result is

$$\begin{aligned}\mathcal{M}_1^{(2)}\big|_{\beta_A^0} &= m_A^2\sum_{k=0}^{3}\widehat{A}_{1k}\,\mathcal{L}^k,\\ \mathcal{M}_2^{(2)}\big|_{\beta_A^0} &= m_A^2\sum_{k=0}^{2}\widehat{A}_{2k}\,\mathcal{L}^k.\end{aligned} \tag{38}$$

The highest logarithmic coefficients are

$$\begin{aligned}&\widehat{A}_{13} = -\tfrac{28}{3}\pi N_c^2, \qquad \widehat{A}_{12} = -\tfrac{14}{3}\pi N_c(11N_c - 2n_f),\\ &\widehat{A}_{22} = -2\pi N_c^2, \qquad \widehat{A}_{21} = -\tfrac{4}{3}\pi N_c(4N_c - n_f),\end{aligned} \tag{39}$$

which do not depend on $\mu_R$, and

$$\widehat{A}_{11} = \frac{\pi N_c}{9}\bigl(N_c\,a_\mathrm{A} - n_f\,a_\mathrm{F}\bigr), \tag{40}$$

where

$$\begin{aligned}a_\mathrm{A} &= 1454 + 462\ln 2 + 252\ln^2 2 - 42\pi^2 + 693\,i\pi - 693L_\mu,\\ a_\mathrm{F} &= 152 + 84\ln 2 + 126\,i\pi - 126L_\mu,\end{aligned}$$

and $L_\mu = \ln(m_A^2/\mu_R^2)$. Table V gives all coefficients for $N_c = 3$, $n_f = 5$ and $\mu_R = m_A$. At $\beta_A^2$, $\beta_A^4$ and $\beta_A^6$ the same substitution again removes every $\ln(1\pm c)$ and $\mathrm{Li}_2$; the remaining dependence on the angle is rational, even in $c$ and free of poles at $c = \pm1$.

Up to the constant $i\pi/2$ and power corrections, $\mathcal{L}$ is the logarithm of the transverse momentum of the pseudoscalars or, equivalently, of the Gram determinant $tu - m_A^4$. The exact one-loop remainders already contain it, since they depend on $\ln(y-x)+\ln(z-x)$ and $(y-x)(z-x) = yz-1 = (tu-m_A^4)/m_A^4$ by Eq. (10). The two-loop coefficients have no finite limit as $\beta_A \to 0$; they grow as $\mathcal{L}^3$ and $\mathcal{L}^2$.

### B. One loop

Fitting the published one-loop remainders at $\beta_A = 10^{-40}$ and $10^{-50}$ with 400-digit arithmetic, for $N_c = 3$, 4 and 5, $n_f = 0$, 2

TABLE V. Coefficients of the leading power term, $\mathcal{M}_i|_{\beta_A^0} = m_A^2 \sum_k \widehat{A}_{ik}\, \mathcal{L}^k$, with $\mathcal{L}$ of Eq. (37). They do not depend on the scattering angle. Exact forms for $\mu_R = m_A$ and general $N_c$, $n_f$; numerical values for $N_c = 3$, $n_f = 5$, $\mu_R = m_A$.

| | $k$ | exact | Re $\widehat{A}_{ik}$ | Im $\widehat{A}_{ik}$ |
|---|---|---|---|---|
| $\mathcal{M}_1$ | 3 | $-\frac{28}{3}\pi N_c^2$ | $-2.63894 \times 10^2$ | 0 |
| $\mathcal{M}_1$ | 2 | $-\frac{14}{3}\pi N_c(11N_c - 2n_f)$ | $-1.01159 \times 10^3$ | 0 |
| $\mathcal{M}_1$ | 1 | Eq. (40) | $3.55129 \times 10^3$ | $4.76702 \times 10^3$ |
| $\mathcal{M}_1$ | 0 | – | $-3.98215 \times 10^2$ | $-4.00569 \times 10^3$ |
| $\mathcal{M}_2$ | 2 | $-2\pi N_c^2$ | $-5.65487 \times 10^1$ | 0 |
| $\mathcal{M}_2$ | 1 | $-\frac{4}{3}\pi N_c(4N_c - n_f)$ | $-8.79646 \times 10^1$ | 0 |
| $\mathcal{M}_2$ | 0 | – | $3.12007 \times 10^2$ | $3.40501 \times 10^2$ |

TABLE VI. Behaviour of the Type II coefficients at threshold, $\beta_A \to 0$ at fixed angle: tree-level values, and the highest power of $\mathcal{L}$ at one and two loops.

| | tree level | one loop | two loops |
|---|---|---|---|
| $\mathcal{M}_1$ ($J_z = 0$) | $2im_A^2$ | $\mathcal{L}$ | $\mathcal{L}^3$ |
| $\mathcal{M}_2$ ($\lvert J_z\rvert = 2$) | $2im_A^2\beta_A^2(1 - c^2)$ | $\mathcal{L}^0$ | $\mathcal{L}^2$ |

and 5, two values of $\mu_R$ and several angles, I find to 30 digits

$$\mathcal{M}_1^{(1)} = m_A^2\big[7\pi N_c\, \mathcal{L} + \widehat{A}_{10}^{(1)}\big] + O(\beta_A),$$
$$\mathcal{M}_2^{(1)} = \tfrac{1}{2}\pi N_c\, m_A^2 + O(\beta_A), \tag{41}$$

with $\widehat{A}_{10}^{(1)} = -93.900 + 11.484\,i$ for $N_c = 3$, $n_f = 5$ and $\mu_R = m_A$, in the normalisation of Eq. (23). The constant does not depend on the angle, and by Eq. (23) its scale dependence is $\partial \widehat{A}_{10}^{(1)} / \partial \ln \mu_R^2 = 4i\beta_0$. The coefficient of $\mathcal{L}$ and the limit of $\mathcal{M}_2^{(1)}$ depend neither on $n_f$ nor on $\mu_R$. Together with Eqs. (19) and (38), the behaviour at threshold is summarised in Table VI. The $|J_z| = 2$ amplitude is suppressed near threshold only at tree level. According to Eq. (18), the helicity amplitude $\mathcal{A}_{\pm\mp}$ then keeps its dependence $e^{\pm 2i\phi}$ on the azimuth of $\vec{p}_3$ as $|\vec{p}_3| \to 0$. An amplitude that is analytic in $\vec{p}_3$ at $\vec{p}_3 = 0$ cannot behave in this way; the loop amplitudes can, because they depend on $\ln p_T$.

### C. The double parton scattering singularity

*a. Origin of the logarithm.* The logarithm $\mathcal{L}$ in Eq. (41) can be understood by examining the Type II diagrams near $p_T = 0$. In the diagram of Fig. 1(b), each incoming gluon splits into two gluons through a three-gluon vertex. Gluons from different incoming partons then fuse pairwise into the pseudoscalars at the two effective vertices. As $p_T \to 0$, all four internal gluons can become simultaneously on shell, with two collinear to each incoming gluon. This configuration is the pinch that produces the DPS singularity of the crossed box [2].

The singular contribution can be expressed as an integral over transverse momentum of the product of the light cone wavefunctions of the two incoming gluons and the tree amplitudes of the two hard subprocesses [2]. The wavefunctions describe the splitting of each incoming parton. The hard subprocess amplitudes are evaluated with collinear on-shell incoming partons and zero transverse momentum. Thus, the splittings determine the strength of the singularity, while the hard subprocesses enter its coefficient. In the present case, the splittings are $g \to gg$ and the subprocesses are $gg \to A$ through $O_G$.

For a gluon splitting into a collinear pair, the daughter helicities add to 0 or $\pm 2$, whereas the parent helicity is $\pm 1$. The splitting amplitude therefore vanishes in the exactly collinear limit. For the allowed tree helicity configurations, its numerator is linear in the relative transverse momentum. After including the propagator, the corresponding light cone wavefunction behaves as the inverse of that momentum. The transverse momentum integral is consequently at most logarithmically singular, as in the massless quark examples of Ref. [2].

*b. Helicity selection and weights.* Whether this logarithm is present also depends on the helicities of the complete singular configuration. For the fermion loops considered in Ref. [2], helicity conservation forces the four intermediate partons to have $J_z = 0$. This restriction does not hold for general gluon loops. The condition that remains is conservation of $J_z$ between the incoming state and the four intermediate partons [3].

For $gg \to AA$, however, each effective vertex requires the two fusing gluons to have equal helicities: the amplitude for $gg \to A$ vanishes for opposite helicities. The gluons in each fusing pair travel in opposite beam directions, so equal helicities give $J_z = 0$. The four intermediate gluons therefore have total $J_z = 0$, and a DPS logarithm is possible only for an incoming state with $J_z = 0$. This explains why the equal helicity coefficient $\mathcal{M}_1$ contains a logarithm at one loop, whereas the opposite helicity coefficient $\mathcal{M}_2$ does not.

To determine the dependence of the logarithmic coefficient on the momentum fractions, consider an incoming gluon of positive helicity splitting into gluons $a$ and $b$. Let their longitudinal momentum fractions be $z$ and $1 - z$. With the coupling, colour and transverse momentum factors suppressed,

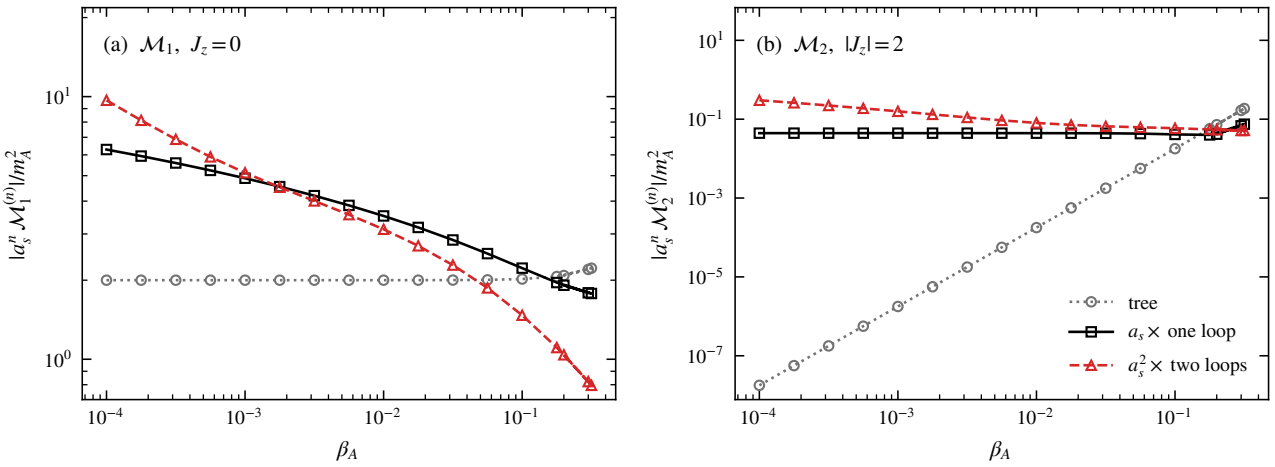



FIG. 3. Tree-level, one-loop and two-loop terms of the Type II amplitude, $|a_s^n \mathcal{M}_i^{(n)}|/m_A^2$ with $a_s = 0.0094$ ($\alpha_s = 0.118$), at $c = 1/3$, $m_A = \mu_R = 125\,\text{GeV}$, $N_c = 3$ and $n_f = 5$. Tree level: Eq. (19); one loop: published remainder; two loops: series through $\beta_A^6$. $\mathcal{M}_1$ and $\mathcal{M}_2$ are the equal and opposite helicity amplitudes, Eq. (18).

the longitudinal weights are

$$w_{++}(z) = \frac{1}{\sqrt{z(1-z)}},$$
$$w_{+-}(z) = \frac{z^2}{\sqrt{z(1-z)}},$$
$$w_{-+}(z) = \frac{(1-z)^2}{\sqrt{z(1-z)}}. \tag{42}$$

The subscripts denote the daughter helicities $(h_a, h_b)$ [40, 41]. These are the longitudinal factors, not the complete splitting amplitudes. The sum of their squares is

$$\frac{1+z^4+(1-z)^4}{z(1-z)}, \tag{43}$$

which gives the momentum fraction dependence of the $g \to gg$ splitting function.

At the DPS point, let the first pseudoscalar receive a fraction $x$ of the first incoming gluon momentum and a fraction $1-x$ of the second. The complementary fractions follow from the equal final state masses, as in Ref. [2]. In Fig. 1(b), this corresponds to $k = xp_1$ and $p_3 - k = (1-x)p_2$. Here $x$ and $z$ denote longitudinal momentum fractions, rather than the kinematic variables of Eq. (10).

The CP properties of the effective vertex fix the relative signs in the helicity sum. Parity reverses all helicities, and since $A$ is odd under parity, the amplitudes for $g^+g^+ \to A$ and $g^-g^- \to A$ have opposite signs [41]. For two positive-helicity incoming gluons, and with the first helicity of each pair referring to the gluon that goes to the first pseudoscalar, the allowed configurations are those in which both splittings produce $(+,+)$, both produce $(+,-)$, or both produce $(-,+)$. In the first configuration both effective vertices receive two positive-helicity gluons. In each mixed configuration, one vertex receives two positive-helicity gluons and the other two negative-helicity gluons. The latter two contributions therefore enter with the opposite sign. Applying the factorised form of Ref. [2] with these weights, their sum gives

$$\begin{aligned} F_A(x) &= w_{++}(x)w_{++}(1-x) \\ &\quad - w_{+-}(x)w_{+-}(1-x) - w_{-+}(x)w_{-+}(1-x) \\ &= \frac{1}{x(1-x)} - 2x(1-x). \end{aligned} \tag{44}$$

Each mixed product contributes $x(1-x)$, which accounts for the second term.

For a CP-even coupling to $G^2$, the two subprocess amplitudes have the same sign. The mixed contributions then add, giving

$$F_S(x) = \frac{1}{x(1-x)} + 2x(1-x). \tag{45}$$

These weights concern gluon loops with two effective vertices. They differ from those of the massless quark boxes for $gg \to HH$ considered in Ref. [2]. For those boxes, the weight is proportional to $x(1-x)$ because internal quarks of equal helicity carry complementary momentum fractions.

*c. Normalisation and the forward limit.* At threshold, both pseudoscalars are at rest and $x = \frac{1}{2}$, so that $F_A(\frac{1}{2}) = \frac{7}{2}$ and $F_S(\frac{1}{2}) = \frac{9}{2}$. By Eq. (37), the coefficient of $\ln p_T^2$ is half the coefficient of $\mathcal{L}$. The one-loop threshold result in Eq. (41) therefore gives

$$\frac{7\pi N_c\, m_A^2}{2} = \frac{7}{2}\pi N_c m_A^2 = \pi N_c m_A^2 F_A\left(\frac{1}{2}\right).$$

Matching at threshold thus fixes the overall factor multiplying $F_A$ to $\pi N_c m_A^2$. This factor is obtained from the calculated amplitude; it has not been derived independently from the factorisation formula.

Having fixed this constant, the predicted dependence on $x$ can be tested away from threshold. The forward limit $c \to 1$ at fixed velocity also gives $p_T \to 0$, but the produced bosons now move along the beam. Write $\beta$ and $m$ for their velocity and mass,

with $\beta = \beta_A$ and $m = m_A$ for $AA$. In this limit, the longitudinal velocity of the first boson is $2x - 1$, so $x = (1+\beta)/2$.

To extract the logarithmic coefficient, I differentiate the real part of the one-loop remainder with respect to $\ln p_T^2$ at fixed $\beta$ and $m$, and then take the forward limit. Dividing by the normalisation fixed above defines

$$D(\beta) \equiv \frac{1}{\pi N_c m^2} \lim_{c\to 1} \frac{\partial\, \mathrm{Re}\, \mathcal{M}_1^{(1)}}{\partial \ln p_T^2} = F_A\left(\frac{1+\beta}{2}\right) \qquad (gg \to AA). \tag{46}$$

The last equality is the prediction to be tested: no further adjustment of the normalisation is made as $\beta$ is varied.

I evaluated $D(\beta)$ from the exact one-loop remainders using $1 - c = 10^{-14}$ and $10^{-18}$, over $0.001 \le \beta \le 0.7$, with $N_c = 3$ and 4 and $n_f = 0$, 2 and 5. The largest relative deviation from Eq. (46) was $3 \times 10^{-15}$. The extracted coefficients follow the pseudoscalar weight $F_A$, rather than the scalar weight $F_S$, over the tested range.

*d. Comparison with scalar pair production.* The scalar process provides a separate test of the relative signs in the helicity sum. The splittings and colour factors are unchanged, while replacing the CP-odd vertices by CP-even ones changes $F_A$ to $F_S$. I tested this prediction using the one-loop Type II amplitudes for $gg \to HH$ at the same mass $m$, computed with the code of Ref. [7] after replacing the effective vertices. The code reproduces the one-loop $AA$ result of that reference to 40 digits, and the $HH$ double poles agree with Catani's formula [30].

In the conventions used here, the $HH$ tree amplitude at threshold is $-2im^2$, opposite in sign to the $AA$ amplitude. The corresponding forward coefficient satisfies

$$D(\beta) = -F_S\left(\frac{1+\beta}{2}\right) \qquad (gg \to HH)$$

to a relative accuracy of $3 \times 10^{-15}$ for $0.001 \le \beta \le 0.7$. The normalisation fixed from $AA$ therefore also reproduces the $HH$ logarithm without an additional fitted constant. In units of the respective tree amplitudes, the ratio of the scalar and pseudoscalar logarithmic coefficients at threshold is

$$\frac{F_S(\frac{1}{2})}{F_A(\frac{1}{2})} = \frac{9}{7}. \tag{47}$$

The opposite helicity amplitude again has no logarithm. Its extracted forward coefficient is below $3 \times 10^{-14}$ times that of $\mathcal{M}_1$. For this check, the diagram containing the four-gluon term of the CP-even operator and the contact counterterm for two such insertions were not included. Neither depends on $p_T$, so neither changes the logarithmic coefficient being tested.

Figure 4 compares the extracted coefficients with the two helicity weights. The $AA$ points for $(N_c, n_f) = (3,5)$ and $(4,2)$ lie on top of each other and on the curve $F_A$, so that after the normalisation of Eq. (46) the coefficient does not depend on $N_c$ or $n_f$. The $HH$ points (triangles), plotted with their sign reversed, lie on the curve $F_S$, which is separated from $F_A$ by

$$F_S - F_A = 4x(1-x) = 1 - \beta^2. \tag{48}$$

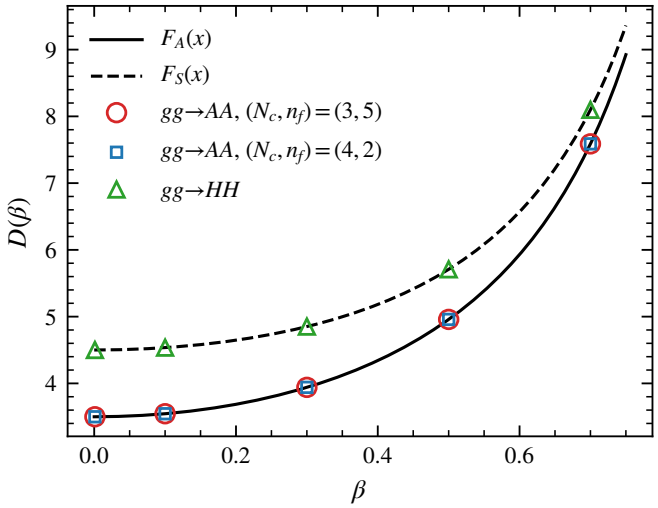



FIG. 4. Comparison of the forward logarithmic coefficients with the helicity weights. Circles and squares show $D(\beta)$ for $gg \to AA$, defined in Eq. (46), for $(N_c, n_f) = (3,5)$ and $(4,2)$, respectively. The circles and squares coincide. Triangles show $-D(\beta)$ for $gg \to HH$ through two CP-even effective vertices at $(N_c, n_f) = (3,5)$. The sign reversal accounts for the relative sign of the scalar and pseudoscalar tree amplitudes at threshold. The solid and dashed curves are $F_A((1+\beta)/2)$ and $F_S((1+\beta)/2)$, respectively.

*e. Colour, scale dependence and phase.* The factorisation also explains the flavour and colour dependence of the one-loop logarithm. The vertices of $O_G$ couple only to gluons, so only gluon splittings enter this singular configuration and its coefficient is independent of $n_f$. The factor $N_c$ comes from the colour contraction of the two three-gluon vertices. Renormalisation and infrared subtraction do not change this logarithmic coefficient: at one loop, their contributions are proportional to the tree amplitude and do not generate a logarithm of $p_T$. The coefficient is therefore independent of $\mu_R$.

Finally, the coefficient $7\pi N_c\, m_A^2$ of $\mathcal{L}$ is real, whereas $\mathcal{M}_1^{(0)} \to 2im_A^2$ at threshold. The logarithm is thus out of phase with the tree amplitude by 90°. This relative phase is consistent with the absorptive origin of the one-loop DPS singularity, which can be obtained from cuts that put internal lines on shell [2]. It explains why the one-loop DPS logarithm does not contribute to the interference with the tree amplitude, as discussed in Sec. V.

### D. Two loops

The leading logarithms of the calculated two-loop coefficient $\mathcal{M}_1$ can be written in terms of the one-loop DPS logarithm. Comparing the coefficients of Eq. (39) with the coefficient $7\pi N_c$ of Eq. (41) gives, for general $N_c$ and $n_f$,

$$\widehat{A}_{13}\mathcal{L}^3 + \widehat{A}_{12}\mathcal{L}^2 = 7\pi N_c\, \mathcal{L}\left[-\tfrac{4N_c}{3}\,\mathcal{L}^2 - 2\beta_0\,\mathcal{L}\right], \tag{49}$$

with $\beta_0$ as in Eq. (23). A similar relation holds in planar $\mathcal{N}=4$ super Yang–Mills theory. Reference [3] studied the six-gluon maximally helicity violating (MHV) amplitude, which has two negative and four positive gluon helicities when all momenta are taken as outgoing. Near the configuration that mimics double parton scattering, its singular terms obey an evolution equation that determines the leading logarithms to all orders. The ratio of the leading two-loop and one-loop terms, including their powers of the coupling, is $-(a/6)\ln^2\delta$, where $a = g^2 N_c/(8\pi^2)$ and $\delta$ measures the approach to the singular configuration; at fixed values of the other kinematic variables, $\delta$ is proportional to the squared transverse momentum. With $a = 2N_c a_s$ and $\ln\delta = 2\mathcal{L} + \text{const}$, the ratio becomes $-\frac{4}{3}N_c a_s \mathcal{L}^2$ at leading logarithmic order. This agrees with the ratio of the cubic term in Eq. (49) to the one-loop logarithm, once the additional power of $a_s$ at two loops is included. The second term, $-2\beta_0\mathcal{L}$, has no counterpart in the $\mathcal{N}=4$ theory, whose $\beta$ function vanishes, and the number of quark flavours enters it only through $\beta_0$.

The same $\beta_0$ term appears in the momentum-space kernel for the splitting $g \to gg$ in unpolarised colour singlet double parton distributions at NLO [22], where the transverse momentum mismatch $\Delta$ between amplitude and conjugate plays the role of $p_T$. Relative to the LO term, the $\beta_0$ part of the coefficient of $\ln^2(\mu^2/\Delta^2)$, given by Eqs. (50) and (135) of Ref. [22] in units of $\alpha_s/2\pi$, corresponds to $-2\beta_0\mathcal{L}$ in the normalisation of $a_s$ used here. One half comes from the renormalisation of the coupling and the other half from the $\delta$-function parts of the DGLAP kernels, so that the running of the coupling alone would give $-\beta_0\mathcal{L}$. The remaining terms, one proportional to $C_A \ln[x(1-x)]$ from the plus distributions, which contributes where the two momentum fractions add up to one, and the real emission terms, which contribute where they add up to less than one, have no counterpart in the amplitude.

The $|J_z| = 2$ amplitude acquires a logarithm at two loops, $\widehat{A}_{22} = (-4N_c)\cdot\frac{1}{2}\pi N_c$, that is the one-loop constant multiplied by $-4N_c\,\mathcal{L}^2$. The same relation holds away from threshold. In the forward limit $c \to 1$ at fixed velocity, where $p_T \to 0$ as well, I evaluated the released two-loop remainder of $\mathcal{M}_2$ at $1-c = 10^{-5}$, $10^{-6}$ and $10^{-7}$ with 140-digit arithmetic, checked with 180 digits, which agree to at least 51 digits, and fitted a quadratic polynomial in $\ln(1-c)$. For $0.05 \le \beta \le 0.5$ and $(N_c, n_f) = (3,5)$ and $(4,2)$, the coefficient of $\ln^2(2p_T/m)$ equals $-4N_c$ times the forward limit of $\mathcal{M}_2^{(1)}$ to $6\times10^{-5}$, while that limit itself varies by a factor of 3.4 over this range. The one-loop amplitude and the two-loop logarithm therefore combine as $a_s\mathcal{M}_2^{(1)}\left[1-4N_c a_s \ln^2(2p_T/m)\right]$, and the logarithm has no velocity or flavour dependence of its own. This factor is the first term of $\exp(-4N_c a_s\mathcal{L}^2)$, which, multiplied by the one-loop coefficient $7\pi N_c$ and integrated over $\mathcal{L}$, gives the cubic term of Eq. (49), and which corresponds at lowest order to the factor $\exp(-\gamma_K \ln^2\delta/8)$ of Ref. [3], with the cusp anomalous dimension $\gamma_K = 4a$ at lowest order; whether it exponentiates in QCD cannot be decided from two loops. The $J_z$ argument of Sec. IV C concerns the one-loop singularity and does not forbid the logarithm. It cannot come from two-loop amplitudes of the subprocesses, since conservation of angular momentum forbids $gg \to A$ for opposite gluon helicities to all orders, and the mechanism that produces it remains open. A related effect occurs in planar $\mathcal{N}=4$ super Yang–Mills theory: the next-to-MHV (NMHV) six-gluon amplitude, with three negative helicities, has no such singularity at one loop because of $J_z$ conservation, but has logarithmic singularities from two loops on, found where the limit overlaps with multi-Regge kinematics; Dixon and Esterlis suggest that they may come from contributions in which more than one particle crosses a cut [3].

Figure 3 compares the three orders. With $a_s = 0.0094$ ($\alpha_s = 0.118$), at $\beta_A = 0.1$ and $c = 1/3$, the one- and two-loop terms of $\mathcal{M}_1$ are 1.1 and 0.7 times the tree term, and the one-loop term of $\mathcal{M}_2$ is 2.3 times the tree term. The last ratio grows as $\beta_A^{-2}$ towards threshold, since the tree-level $|J_z| = 2$ amplitude vanishes there. The one-loop terms include the scale term of Eq. (23). Towards threshold the loop terms of $\mathcal{M}_1$ grow with $|\mathcal{L}|$, the two-loop term as $|\mathcal{L}|^3$ and the one-loop term as $|\mathcal{L}|$, so that below $\beta_A \approx 1.7\times10^{-3}$ the two-loop term exceeds the one-loop term at this angle. Because the logarithms are out of phase with the tree amplitude, this growth largely drops out of the cross section (Sec. V). In panel (b) the one-loop term of $\mathcal{M}_2$ tends to the constant of Eq. (41), while the two-loop term grows as $\mathcal{L}^2$, the logarithm that the $|J_z| = 2$ amplitude acquires at two loops.

## V. CONSEQUENCES FOR THE CROSS SECTION

### A. Leading order

Averaging Eq. (18) over helicities and colours, and including a factor $\frac{1}{2}$ for identical particles, gives the LO partonic cross section of the Type II amplitude,

$$\frac{\mathrm{d}\hat{\sigma}}{\mathrm{d}c} = \frac{\beta_A\, C_G^4}{128\pi\hat{s}\,(N_c^2-1)}\left(|\mathcal{M}_1^{(0)}|^2 + |\mathcal{M}_2^{(0)}|^2\right) \qquad (50)$$

for $-1 < c < 1$. For $\alpha_s = 0.118$, $\cot\beta = 1$ and $m_A = 125\,\text{GeV}$ I find $\hat{\sigma} = 1.29\times10^{-8}\,\text{pb}$ at $\beta_A = 0.1$ (Fig. 5(a)). The cross section rises as $\beta_A$ above threshold and as $\hat{s}$ at large $\hat{s}$, as expected for dimension-five operators, and it is of fourth order in $a_s$ through $C_G^4$. In a 2HDM, it is a small part of the total, which is dominated by Type I (Sec. II E).

### B. Finite virtual corrections

To examine the effect of the loop corrections on the Type II squared amplitude, I define two dimensionless coefficients, $V_1$ and $V_2$, relative to the tree contribution. Let $\mathcal{A}_{\text{fin}}$ denote the amplitude constructed from the finite remainders of Eq. (21), in the normalisation of Eq. (23). Using the helicity decomposition

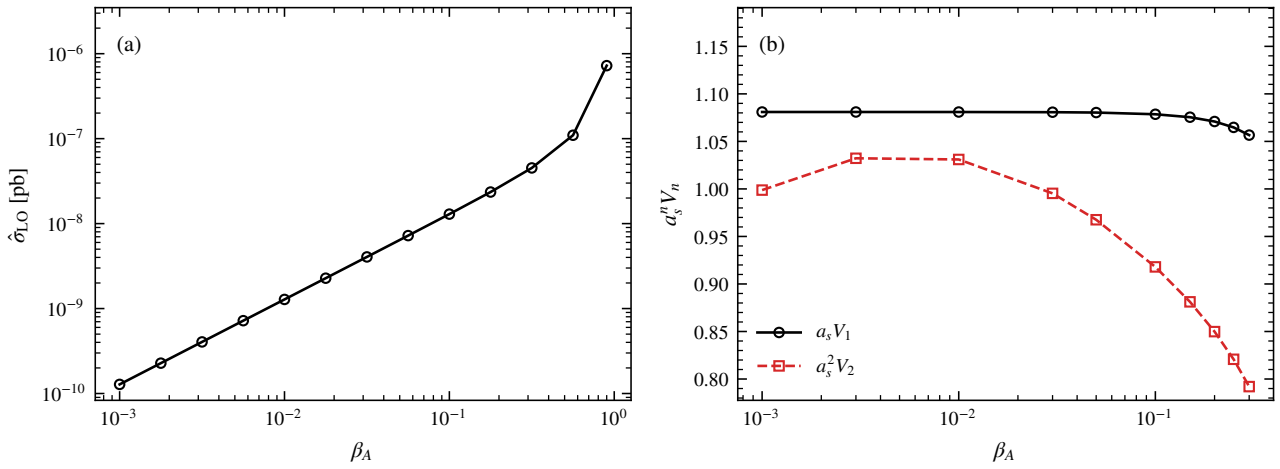



FIG. 5. (a) Leading order partonic cross section of the Type II amplitude, Eq. (50), for $m_A = 125\,\text{GeV}$, $\alpha_s = 0.118$ and $\cot\beta = 1$. (b) Angle averages $a_s\langle V_1\rangle$ and $a_s^2\langle V_2\rangle$ of the finite virtual corrections, Eq. (52), for $\mu_R = m_A$, $N_c = 3$ and $n_f = 5$.

in Eq. (18), its square can be written as

$$\sum_{\text{hel,col}} |\mathcal{A}_{\text{fin}}|^2 = 2(N_c^2-1)\,C_G^4 \sum_i |\mathcal{M}_i^{(0)}|^2 \times \left[1 + a_s V_1 + a_s^2 V_2 + O(a_s^3)\right],$$
$$V_1 = \frac{2\,\text{Re}\sum_i \mathcal{M}_i^{(0)*}\mathcal{M}_i^{(1)}}{\sum_i |\mathcal{M}_i^{(0)}|^2},$$
$$V_2 = \frac{\sum_i \left(2\,\text{Re}\left[\mathcal{M}_i^{(0)*}\mathcal{M}_i^{(2)}\right] + |\mathcal{M}_i^{(1)}|^2\right)}{\sum_i |\mathcal{M}_i^{(0)}|^2}. \qquad (51)$$

The sums run over the two tensor coefficients, $i = 1, 2$. The coefficient $V_1$ contains the interference between the tree and one-loop amplitudes. At the next order, $V_2$ contains both the interference between the tree and two-loop amplitudes and the square of the one-loop amplitude.

These coefficients refer only to the $O_G$–$O_G$ Type II contribution. Their interpretation as NLO and NNLO finite virtual terms applies when the Type II tree amplitude is the leading amplitude, as for a pseudoscalar coupled to gluons only through $G\tilde{G}$ (Sec. II E). They do not include interference with the Type I amplitudes present in a 2HDM.

The quantities $V_1$ and $V_2$ depend on the subtraction scheme of Eq. (21); they are not observables or $K$ factors. A physical cross section requires the virtual and real emission contributions to be combined consistently with the counterterms for initial state collinear factorisation. The real emission contributions are not included here. At fixed $\beta_A$, both coefficients depend on the scattering angle. To display their velocity dependence in Fig. 5(b), I average them over the angle using the Type II LO cross section as the weight:

$$\langle V_n\rangle = \frac{\displaystyle\int_{-1}^{1} \mathrm{d}c \sum_i |\mathcal{M}_i^{(0)}|^2\, V_n}{\displaystyle\int_{-1}^{1} \mathrm{d}c \sum_i |\mathcal{M}_i^{(0)}|^2}, \qquad n = 1, 2. \qquad (52)$$

At fixed $\beta_A$, the remaining factors in Eq. (50) are independent of $c$ and cancel in this ratio. Thus, $a_s\langle V_1\rangle$ and $a_s^2\langle V_2\rangle$ give the finite virtual contributions relative to the angularly integrated tree contribution, rather than an unweighted average of the corrections. The integrals are evaluated with eight-point Gauss–Legendre quadrature in $c$; with 32 points, $\langle V_1\rangle$ changes by less than $10^{-9}$ and $\langle V_2\rangle$ by less than $10^{-3}$ relative to its value at $\beta_A = 0.01$, 0.1 and 0.3.

I evaluate the one-loop remainders without a threshold expansion and use the series through $\beta_A^6$ for the two-loop remainders. For $\alpha_s = 0.118$ and $\mu_R = m_A = 125\,\text{GeV}$, $a_s\langle V_1\rangle$ lies between 1.06 and 1.08, while $a_s^2\langle V_2\rangle$ lies between 0.79 and 1.03 over $10^{-3} \le \beta_A \le 0.3$. These are the two curves shown in Fig. 5(b). At fixed kinematics, the scale dependence of the one-loop remainder in Eq. (23) implies $\partial V_1/\partial \ln\mu_R^2 = 4\beta_0$.

*a. One-loop contribution.* The threshold behaviour of $V_1$ and $V_2$ follows from the relative phases of the tree and loop amplitudes. At fixed scattering angle, $\mathcal{M}_1^{(0)} \to 2im_A^2$, whereas $\mathcal{M}_2^{(0)}$ vanishes as $\beta_A^2$. The squared tree amplitude in the denominator of Eq. (51) therefore tends to $4m_A^4$. At leading power, interference with the tree amplitude selects the imaginary part of $\mathcal{M}_1^{(1)}$ or $\mathcal{M}_1^{(2)}$.

To separate the real logarithm from the constant phase, define

$$\ell = \ln\frac{2p_T}{m_A}, \qquad \mathcal{L} = \ell + \frac{i\pi}{2} + O(\beta_A^2),$$

where the second relation follows from Eq. (37). At leading power, the one-loop result in Eq. (41) is

$$\frac{\mathcal{M}_1^{(1)}}{m_A^2} = 7\pi N_c\,\ell + \frac{i\pi}{2}7\pi N_c + \widehat{A}_{10}^{(1)}.$$

The coefficient of $\ell$ is real, so this term does not interfere with the imaginary tree amplitude. The imaginary constant terms do contribute, giving

$$V_1 \longrightarrow \frac{\pi}{2}7\pi N_c + \text{Im}\,\widehat{A}_{10}^{(1)}. \qquad (53)$$

For $N_c = 3$, $n_f = 5$ and $\mu_R = m_A$, this gives $V_1 \to 115.11$. Thus, the one-loop remainder contains a DPS logarithm, but the leading power finite virtual correction does not. Its limiting value is independent of the angle, which explains why the angular average in Fig. 5(b) is nearly constant.

*b. Two-loop contribution.* At the next order, the interference of the two-loop amplitude with the tree amplitude again selects the imaginary part of the loop remainder. The square of the one-loop amplitude, however, retains the real logarithm. Using Eqs. (38) and (41), the leading power terms of Eq. (51) can be written as

$$V_2\big|_{\beta_A^0} = \mathrm{Im}\left[\sum_{k=0}^{3} \widehat{A}_{1k}\mathcal{L}^k\right] + \frac{1}{4}\left|7\pi N_c\,\mathcal{L} + \widehat{A}_{10}^{(1)}\right|^2 + \frac{\pi^2 N_c^2}{16}. \tag{54}$$

The first term is the interference of the two-loop amplitude with the tree amplitude, and the second is the square of $\mathcal{M}_1^{(1)}$. The last term comes from $|\mathcal{M}_2^{(1)}|^2$, since $\mathcal{M}_2^{(1)} \to \pi N_c m_A^2/2$. It contributes at leading power even though the corresponding tree amplitude vanishes. By contrast, interference between $\mathcal{M}_2^{(0)}$ and $\mathcal{M}_2^{(2)}$ is suppressed by $\beta_A^2$, up to logarithms.

The coefficients $\widehat{A}_{13}$ and $\widehat{A}_{12}$ are real. At leading power,

$$\mathrm{Im}\,\mathcal{L}^3 = \frac{3\pi}{2}\ell^2 - \frac{\pi^3}{8}, \qquad \mathrm{Im}\,\mathcal{L}^2 = \pi\ell.$$

Consequently, the cubic logarithm of the two-loop amplitude gives a quadratic, not a cubic, logarithm in its interference with the tree amplitude. The term $\widehat{A}_{12}\mathcal{L}^2$ contributes only a linear logarithm to this interference. The square of the one-loop amplitude is also at most quadratic in $\ell$.

The leading threshold expansion of $V_2$ therefore has the form

$$V_2 = v_0 + v_1\ell + v_2\ell^2 + O_{\log}(\beta_A^2). \tag{55}$$

Here $v_0$, $v_1$ and $v_2$ are the coefficients of the constant, linear and quadratic terms in $\ell$. They all belong to the same perturbative coefficient $V_2$; their subscripts do not count loops.

The coefficient $v_2$ receives two contributions. The interference of the two-loop amplitude with the tree amplitude gives $\frac{3}{2}\pi\widehat{A}_{13} = -14\pi^2 N_c^2$, while the square of the one-loop amplitude gives $\frac{1}{4}(7\pi N_c)^2 = \frac{49}{4}\pi^2 N_c^2$. Their sum is

$$v_2 = -14\pi^2 N_c^2 + \frac{49}{4}\pi^2 N_c^2 = -\frac{7}{4}\pi^2 N_c^2. \tag{56}$$

The two terms partly cancel, with the negative interference contribution larger in magnitude. At leading power, the DPS logarithm thus first appears in the finite virtual correction through $V_2$, where its highest power is $\ell^2$.

A fit of the numerical values of $V_2$ at $c = 1/3$ over $10^{-8} \le \beta_A \le 10^{-6}$ gives $v_2 = -155.446269$. Equation (56) gives $-155.446269$ for $N_c = 3$; the relative difference before rounding is $3 \times 10^{-11}$. For $a_s = 0.0094$, the quadratic term in the finite virtual correction is $a_s^2 v_2 \ell^2 = -0.0137\,\ell^2$.

*c. Remaining coefficients and numerical behaviour.* At leading power, $v_0$, $v_1$ and $v_2$ are independent of the scattering angle. The angular dependence of $V_2$ itself remains through $\ell$. Fits for $v_0$ and $v_1$ at $c = 0$, $\frac{1}{3}$ and $\frac{2}{3}$ agree within a relative spread of $5 \times 10^{-11}$. The coefficient of the linear logarithm follows from the same expansion:

$$\begin{aligned} v_1 &= \pi\widehat{A}_{12} + \mathrm{Im}\,\widehat{A}_{11} + \frac{7\pi N_c}{2}\,\mathrm{Re}\,\widehat{A}_{10}^{(1)} \\ &= \frac{7}{2}\pi N_c\,\mathrm{Re}\,\widehat{A}_{10}^{(1)} + 7\pi^2 N_c\beta_0. \end{aligned}$$

For the benchmark parameters, this gives $v_1 = -1508.5$, in agreement with the numerical fit to a relative difference of $1 \times 10^{-10}$. The coefficient is independent of $\mu_R$, since $\mathrm{Re}\,\widehat{A}_{10}^{(1)}$ is scale independent. The same is true of $v_2$. At leading power, the explicit scale dependence of $V_2$ is therefore confined to $v_0$, hich changes from 8118 to 15421 when $\mu_R$ is increased from $m_A$ to $2m_A$.

Although $\ell^2$ is the highest power of the logarithm, it need not give the largest contribution at the velocities considered. At $c = 1/3$, the ratio $(v_1\ell + v_2\ell^2)/V_2$ is approximately 20% at $\beta_A = 0.1$ and 30% at $\beta_A = 0.01$. The quadratic term alone contributes $-4$% and $-21$% of the total $V_2$, respectively. Since $\ell < 0$ in this region and both $v_1$ and $v_2$ are negative, the linear term is positive and the quadratic term is negative. This competition also explains the shape of the correction as a function of velocity. The leading power expression has a maximum at

$$\ell_{\max} = -\frac{v_1}{2v_2} = -4.85.$$

At $c = 1/3$, this corresponds to $\beta_A \simeq 0.0041$. For larger $\beta_A$, $\ell$ increases and the leading power value of $V_2$ decreases. The same balance between the linear and quadratic terms explains the turnover of the angular average $a_s^2\langle V_2\rangle$ in Fig. 5(b). At $c = 1/3$ and $\beta_A = 0.1$, including the terms beyond leading power changes $V_2$ by only 0.3%.

*d. DPS interpretation and subtraction dependence.* The two contributions to $v_2$ also differ in their relation to the DPS configuration. In the square of the one-loop amplitude, the DPS region occurs in both the amplitude and its complex conjugate, that is, on both sides of the final state cut. This is the configuration associated with perturbative splittings in both protons that overlaps with DPS and requires a subtraction to avoid double counting [23, 27]. In the interference with the tree amplitude, the DPS region occurs only on the loop side of the cut. Both contributions enter the finite virtual coefficient $V_2$ defined in Eq. (51); the logarithmic term in its one-loop square alone would therefore not give the complete coefficient $v_2$.

Although $V_2$ depends on the IR subtraction convention, $v_2$ is unchanged by finite scheme changes that multiply lower order amplitudes by angle independent functions regular at the pair threshold. The tree amplitude has no logarithm, and the one-loop remainder contains at most one power of $\ell$. Such changes can therefore affect the constant and linear terms in $V_2$, but not the coefficient of $\ell^2$.

Finally, with the Wilson coefficient counting specified in Sec. II E, the one-loop $O_G$–$O_J$ amplitude contributes at the

same order as the two-loop $O_G$–$O_G$ amplitude. Its coefficient $C_J$ depends on $m_t$ in the heavy top quark theory [7]. By Eq. (24), the mixed amplitude is proportional to the tree amplitude. After including its Wilson coefficient and matching the normalisation, its interference with the tree amplitude therefore gives a kinematically constant shift to $V_2$. It changes $v_0$, but not $v_1$ or $v_2$, and is not included in the results above.

### C. Use in higher order calculations

The threshold series provide a compact representation of the two-loop Type II finite remainders near the pair production threshold. For the two-body virtual amplitude, $m_{AA} = \sqrt{\hat{s}}$, and

$$\frac{m_{AA}}{2m_A} = \frac{1}{\sqrt{1-\beta_A^2}}.$$

The range $\beta_A \leq 0.2$ therefore corresponds to pair invariant masses up to 2.06% above $2m_A$. For $m_A = 125\,\text{GeV}$, the upper limit is $m_{AA} = 255.2\,\text{GeV}$. The point $\beta_A = 0.3$ lies 4.83% above threshold.

I recommend the series through $\beta_A^6$ for $0 < \beta_A \leq 0.2$ at the angles tested in Tables II and IV. At $\beta_A = 0.3$, the largest relative error of the complete coefficients on the validation grid is $1.94 \times 10^{-3}$. These numerical checks do not establish a uniform accuracy bound as $c \to \pm 1$, where the logarithms diverge. The series retain the complex values of both tensor coefficients, including the phases needed for interference with other amplitudes. Such contributions must be combined using the same normalisation, Wilson coefficient conventions, tensor basis and infrared subtraction scheme.

One application is the calculation of the hard function for a threshold resummed or soft-virtual prediction of the pair invariant mass distribution. In this context, threshold refers to the limit in which little energy is available for additional QCD radiation at fixed $m_{AA}$, rather than to the pair production limit $\beta_A \to 0$ [16, 17, 19]. At leading power in the soft limit, the soft and collinear factors are universal for gluon fusion into a colour singlet final state. The process dependence enters through the finite virtual contributions to the hard function [18, 20]. The hard function is therefore needed at each pair invariant mass, and the present series provide its two-loop Type II amplitude input near $m_{AA} = 2m_A$.

To construct this contribution, the finite remainders must first be combined with the appropriate lower order amplitudes. When the Type II tree amplitude is the leading amplitude, the relevant $O_G$–$O_G$ combinations are those in Eq. (51). In particular, the NNLO virtual term contains both the interference of the two-loop amplitude with the tree amplitude and the square of the one-loop amplitude. Integrated over the scattering angle at fixed $m_{AA}$ and divided by the integrated tree contribution, they give the averages of Eq. (52). The finite virtual terms must then be combined with the universal contributions according to the subtraction and resummation conventions of the chosen calculation [18]. For Higgs pairs, Ref. [19] gives an explicit construction in which the two-loop finite virtual contribution is integrated over $t$ at fixed pair invariant mass.

The perturbative order of the present contribution depends on the leading amplitude, as explained in Sec. II E. For a pseudoscalar coupled to gluons only through $G\tilde{G}$, the two-loop Type II remainders enter the NNLO virtual contribution. In a 2HDM, they first contribute at N$^3$LO through interference with the leading Type I amplitude. In that case, the Type II angular averages of Eq. (52) cannot be used directly: the interference with Type I must be formed before the angular integration. The remaining amplitudes and any mixed $O_G$–$O_J$ contribution must also be included at the order required by the theory. For Higgs pairs, likewise, the NNLO virtual corrections contain the diagrams with two effective vertices only at tree level and at one loop [42], and their two-loop terms are an ingredient of N$^3$LO [9].

## VI. CONCLUSIONS

I have expanded the two-loop Type II finite remainders for $gg \to AA$ with two $O_G$ insertions about the pair production threshold through $\beta_A^6$ at fixed scattering angle. After the endpoint constants are expressed in a basis of eight transcendental constants, all inverse powers of $\beta_A$ and the odd powers through $\beta_A^5$ cancel identically. Every coefficient is even in $\cos\theta$, as required by Bose symmetry. For $0 < \beta_A \leq 0.2$, the maximum relative errors of the two complete coefficients are $3.23 \times 10^{-6}$ and $6.77 \times 10^{-5}$, respectively, at the points tested, including the forward angles. The released series files are a factor of 5061 smaller than the expressions released with Ref. [7]. The timing comparisons in App. D also show faster evaluation for the representations tested.

Comparing the perturbative orders required converting the remainders released with Ref. [7] to a common normalisation. The two-loop files carry an overall factor of 64 relative to the tree and one-loop amplitudes, and the one-loop files are given at $\mu_R = 1$ GeV. After the scale dependence is added and all orders are brought to a common normalisation, the remainders satisfy the renormalisation group relations in Eqs. (B1) and (B3) within the accuracy of the checks described in App. B. The pole cancellation reported in Ref. [7] is unaffected, since all terms of the two-loop subtraction carry the same overall factor.

At leading power, the angular dependence enters entirely through the logarithm of the transverse momentum of the pseudoscalars. I identified the one-loop logarithm with the DPS singularity studied in Ref. [2]. The gluon splitting weights and the relative signs of the CP-odd subprocess amplitudes determine the momentum fraction dependence of its coefficient. With the overall constant fixed at threshold, the resulting expression reproduces the velocity dependence of the exact one-loop coefficient in the forward limit to a relative accuracy of $3 \times 10^{-15}$ over the tested range. The same analysis explains the absence of a one-loop logarithm in the $|J_z| = 2$ amplitude, the independence of the logarithmic coefficient from $n_f$ and $\mu_R$, and its phase relative to the tree amplitude. An explicit one-loop calculation of $gg \to HH$ through two CP-even vertices confirms the corresponding scalar weight $F_S$, using the normalisation fixed from $AA$ and accounting for the relative

sign of the tree amplitudes. The opposite helicity amplitude again has no one-loop DPS logarithm.

At two loops, the $J_z = 0$ amplitude grows as $\ln^3 p_T$ and the $|J_z| = 2$ amplitude as $\ln^2 p_T$. The ratio of the leading two-loop and one-loop logarithms in the $J_z = 0$ amplitude agrees with the corresponding ratio in the all-orders result for planar $\mathcal{N} = 4$ super Yang–Mills theory [3]. The coefficient of the quadratic logarithm in this amplitude is proportional to $\beta_0$, with the structure discussed in Sec. IV D in relation to the NLO $g \to gg$ splitting kernel for double parton distributions [22]. The finite virtual corrections are less singular than the amplitudes. At leading power, the one-loop DPS logarithm does not contribute to $V_1$, because it is out of phase with the tree amplitude. The cubic logarithm is absent from $V_2$, whose highest logarithmic term is

$$V_2 \supset -\frac{7}{4}\pi^2 N_c^2 \ln^2\left(\frac{2p_T}{m_A}\right).$$

This coefficient results from a partial cancellation between the interference of the two-loop amplitude with the tree amplitude and the square of the one-loop amplitude.

The overall factor $\pi N_c m^2$ multiplying the one-loop helicity weight remains to be derived directly from the factorisation formula with the effective vertices. Extending the analysis beyond one loop would help explain the two-loop logarithms. In the forward limit, the two-loop logarithm of the $|J_z| = 2$ amplitude is the one-loop amplitude multiplied by $-4N_c a_s \ln^2(2p_T/m_A)$ at every velocity tested, with the coefficient of the double logarithm that also governs $\mathcal{M}_1$; the mechanism that produces it remains to be identified. The known two-loop amplitudes for $gg \to HH$ through two effective vertices [9] also permit a comparison with the scalar case. Finite top quark mass effects are a separate question. If the one-loop DPS factorisation applies to the full theory, the on-shell triangle form factors in Eq. (26) multiply the logarithmic coefficient by 1.096 at $m_A = 125$ GeV. This estimate does not determine the mass dependence of the remaining one-loop terms or of the two-loop amplitude.

The series provide the Type II two-loop amplitude input to the hard function for the pair invariant mass distribution near $m_{AA} = 2m_A$. They enter at NNLO when the Type II tree amplitude is the leading amplitude, as for a pseudoscalar coupled to gluons only through $G\tilde{G}$. In a 2HDM, they first enter at N$^3$LO through interference with the leading Type I amplitude. A soft-virtual prediction requires these remainders to be combined with the appropriate lower order amplitudes, Wilson coefficients and universal soft and collinear contributions [18, 20]. The mixed $O_G$–$O_J$ contribution must also be included where required by the effective theory and its perturbative counting. This contribution was omitted from the numerical results for $V_2$; it gives a kinematically constant shift and does not change the logarithmic coefficients. The series, the evaluator, the validation records and the code of the construction are provided with the paper [28].

## DATA AVAILABILITY

The threshold series, construction files, validation records, the code of the construction and validation, the evaluator, the one-loop $gg \to HH$ coefficients of Sec. IV C and the scripts that reproduce all tables and figures form a self-contained package. Version 7.2.1 is provided as ancillary files with the arXiv version of this article and as Supplemental Material [28]; the package is maintained at `https://github.com/abhattacharya1992/ggAA-threshold`. The unexpanded finite remainders used as input were released as ancillary files with Ref. [7]; the package downloads them from arXiv and verifies their checksums. The code used for the one-loop $gg \to HH$ amplitudes of Sec. IV C is not included; it is available from the author on request.

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### Appendix A: Endpoint constants

The boundary values of the expansion contain the constants

$$\mathrm{MB}[p,q,a,w] = \frac{a}{(p-1)!}\int_0^1 \mathrm{d}t\,\frac{[-\ln t]^{p-1}}{1-at}\,\phi_q(wt), \quad \text{(A1)}$$

with $\phi_1(z) = -\ln(1-z)$ and $\phi_q(z) = \mathrm{Li}_q(z)$ for $q \geq 2$. Besides the elementary values $\mathrm{MB}[2,2,1,1] = \pi^4/120$ and $\mathrm{MB}[2,1,1,1] = \zeta_3$, thirteen distinct constants occur. They were evaluated by quadrature at high precision, independently of the amplitude data, and reduced to the basis of Eq. (31) with the PSLQ algorithm [38] at 110 digits. Table VII lists the results; the largest residual is $4\times10^{-107}$, and the same relations were obtained in Mathematica.

### Appendix B: Normalisation of the released remainders

Five checks establish the normalisation of Eq. (23) for the tree, one-loop and two-loop amplitudes. They use the tree amplitudes and the one- and two-loop remainders released with Ref. [7], together with the Mathematica code of the original calculation.

*Tree level.* The projections of the two tree diagrams from the original calculation give Eq. (19), after removing a colour factor $N_c^2 - 1$ that is included in the file for $T_1$ but not in the file for $T_2$. A direct evaluation of the two diagrams with the vertex $iC_G\varepsilon_{\mu\alpha\rho\beta}k_1^\mu k_2^\rho \delta^{ab}$, which follows from Eq. (1), gives $|\mathcal{A}_{\pm\pm}| = C_G^2|\mathcal{M}_1^{(0)}|$ and $|\mathcal{A}_{\pm\mp}| = C_G^2|\mathcal{M}_2^{(0)}|$ to 14 digits.

*Tree and one loop.* The coefficient of $n_f$ in $\mathcal{M}_i^{(1)}$, divided by $\mathcal{M}_i^{(0)}$, is the same for $i = 1, 2$ at threshold, and its imaginary

TABLE VII. Reductions of the endpoint constants of Eq. (A1), found with PSLQ at 110 digits; the largest residual is $4 \times 10^{-107}$.

| $[p, q, a, w]$ | reduction |
|---|---|
| $[1, 2, -1, 1]$ | $-\frac{\pi^2 \ln 2}{6} + \frac{5\zeta_3}{8}$ |
| $[1, 2, 1/2, 1/2]$ | $-\frac{\zeta_3}{4} - \frac{\ln^3 2}{6} + \frac{\pi^2 \ln 2}{12}$ |
| $[1, 2, 1/2, 1]$ | $-\zeta_3 + \frac{\pi^2 \ln 2}{4}$ |
| $[2, 1, -1, -1]$ | $\frac{\zeta_3}{8}$ |
| $[2, 1, -1, 1]$ | $-\frac{13\zeta_3}{8} + \frac{\pi^2 \ln 2}{4}$ |
| $[2, 1, 1, -1]$ | $-\frac{\pi^2 \ln 2}{4} + \zeta_3$ |
| $[2, 1, 1, 1/2]$ | $-\frac{5\zeta_3}{8} + \frac{\pi^2 \ln 2}{6}$ |
| $[2, 1, 1/2, 1/2]$ | $-\frac{\ln^3 2}{6} + \frac{\zeta_3}{8}$ |
| $[2, 1, 1/2, 1]$ | $-\frac{\pi^2 \ln 2}{4} + \frac{13\zeta_3}{8}$ |
| $[2, 2, 1, -1]$ | $-\frac{71\pi^4}{1440} - \frac{\pi^2 \ln^2 2}{6} + \frac{\ln^4 2}{6} + 4\,\mathrm{Li}_4(\frac{1}{2}) + \frac{7 \ln 2 \zeta_3}{2}$ |
| $[2, 2, 1, 1/2]$ | $-\mathrm{Li}_4(\frac{1}{2}) - \frac{\ln 2 \zeta_3}{4} - \frac{\pi^2 \ln^2 2}{24} - \frac{\ln^4 2}{24} + \frac{19\pi^4}{1440}$ |
| $[2, 2, 1/2, 1/2]$ | $-\frac{\pi^2 \ln^2 2}{24} + \frac{\ln^4 2}{24} + \frac{\pi^4}{1440} + \frac{\ln 2 \zeta_3}{4}$ |
| $[2, 2, 1/2, 1]$ | $-3\,\mathrm{Li}_4(\frac{1}{2}) - \frac{\pi^2 \ln^2 2}{8} - \frac{\ln^4 2}{8} + \frac{7\pi^4}{288}$ |

part equals $-2\pi/3$ to 15 digits at every tested point. This is the value that the subtraction term $2I^{(1)}\mathcal{M}^{(0)}$ of Eq. (21) produces through the $n_f$ part of $\beta_0$ multiplying $\ln(-\hat{s}/\mu_R^2)$; the one-loop diagrams proportional to $n_f$ have no cut in the $\hat{s}$ channel. This check fixes the normalisation and phase of the tree amplitude relative to the one-loop remainders.

*Scale dependence at one loop.* Apart from the overall factor $m_A^2$, a remainder with the dimension of $m_A^2$ can depend on $m_A$ and $\mu_R$ only through their ratio. The released one-loop remainders do not have this property: when $m_A$ and $\mu_R$ are multiplied by a common factor $\lambda$ at fixed $\beta_A$ and $c$, $\mathcal{M}^{(1)}_{i,\mathrm{publ}}/m_A^2$ changes by $-2\beta_0 \ln\lambda^2\, \mathcal{M}_i^{(0)}/m_A^2$, to a relative accuracy of $7 \times 10^{-56}$. They are independent of $\mu_R$, since their eight $\ln\mu_R^2$ terms cancel. In the original calculation, the amplitude is renormalised with $Z_{GG}^2$, where $Z_{GG} = Z_{a_s}$, and $C_G \propto a_s$. With the subtraction operators of Eq. (21), whose scale dependence enters through $(\mu_R^2/(-\hat{s}))^\epsilon$, renormalisation group invariance of Eq. (20) then requires

$$\frac{\partial \mathcal{M}^{(1)}}{\partial \ln \mu_R^2} = 2\beta_0\, \mathcal{M}^{(0)}. \tag{B1}$$

The first line of Eq. (23) adds this dependence, and the resulting remainders are invariant under the rescaling to $6 \times 10^{-57}$.

*Scale dependence at two loops.* The same ingredients determine the scale dependence at two loops. The subtraction operator $I^{(2)}$ of the original code differs from Eq. (3.4) of Ref. [7] in one respect: it does not multiply the constant

$$H_g^{(2)} = \tfrac{1}{2}\Big[C_A^2\big(-\tfrac{5}{12} - \tfrac{11}{24}\zeta_2 - \tfrac{1}{2}\zeta_3\big) + C_A n_f\big(\tfrac{29}{27} + \tfrac{\zeta_2}{12}\big) - \tfrac{1}{2}C_F n_f - \tfrac{5}{27}n_f^2\Big] \tag{B2}$$

by $(-\hat{s}/\mu_R^2)^\epsilon$. With this operator, one finds

$$\frac{\partial \mathcal{M}^{(2)}}{\partial \ln \mu_R^2} = 3\beta_0\, \mathcal{M}^{(1)} + \big(2\beta_1 - 8H_g^{(2)} - 12 C_F n_f\big)\mathcal{M}^{(0)}, \tag{B3}$$

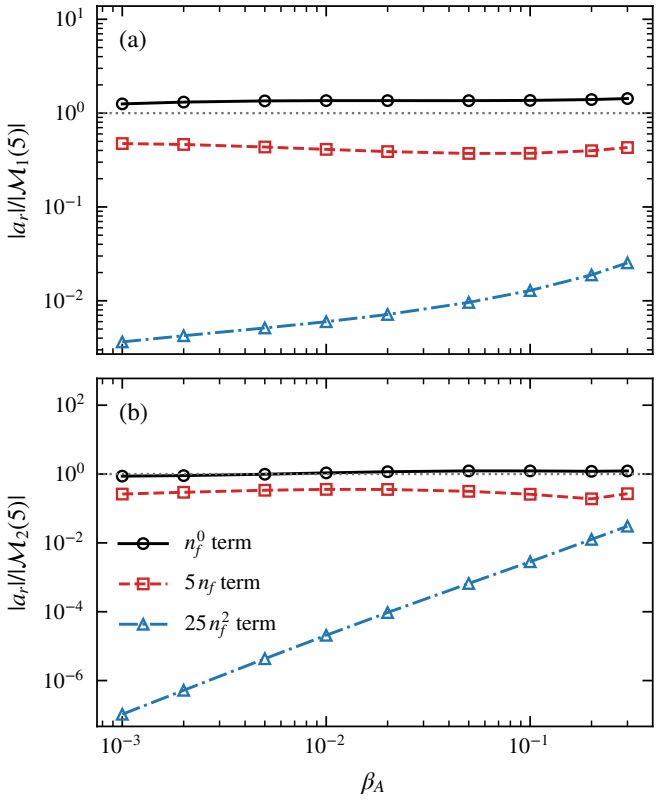



FIG. 6. Magnitudes of the three terms of Eq. (C1), divided by the magnitude of the complete coefficient: (a) $\mathcal{M}_1$, (b) $\mathcal{M}_2$, for $c = 1/3$, $m_A = \mu_R = 125\,\mathrm{GeV}$ and $N_c = 3$. The horizontal line marks unity.

where $\beta_1 = \frac{34}{3}C_A^2 - 2C_F n_f - \frac{10}{3}C_A n_f$. The last term comes from the counterterm of the $O_G$–$O_J$ amplitude, Eq. (24), in the renormalisation of Ref. [7]. I fitted $\partial\mathcal{M}^{(2)}/\partial\ln\mu_R^2 = a\,\beta_0\mathcal{M}^{(1)} + b\,\mathcal{M}^{(0)}$ to the series at two kinematic points, for $(N_c, n_f, m_A) = (3, 0, 300\,\mathrm{GeV})$, $(3, 5, 125\,\mathrm{GeV})$ and $(4, 2, 150\,\mathrm{GeV})$. With both lines of Eq. (23), the fit gives $a = 3$ to $2 \times 10^{-7}$, and $b$ agrees with Eq. (B3) to $2 \times 10^{-5}$, the accuracy of the series at these velocities. The second derivative equals $6\beta_0^2\mathcal{M}^{(0)}$, the value implied by Eqs. (B1) and (B3), to $1 \times 10^{-8}$. Without the division by 64, the fit gives $a = 192$. Without the scale term of the one-loop remainders, $b$ differs from Eq. (B3) by $6\beta_0^2 \ln(m_A^2/\mathrm{GeV}^2)$. The agreement therefore also fixes the reference scale of the released one-loop remainders at $\mu_R = 1\,\mathrm{GeV}$.

*Origin of the factor 64.* The Mathematica code of the original calculation documents this factor. The two-loop renormalisation step multiplies the $O_G$–$O_G$ amplitude by $64 = 8^2$, including the tree and one-loop amplitudes that enter the infrared subtraction, whereas the one-loop step does not. The pole cancellation of Ref. [7] is unaffected, since all terms of the two-loop subtraction carry the same factor. The code and data released with this article use the normalisation of Eq. (23).

*The original output.* The Mathematica output of the original calculation confirms both conversions. Its one-loop remainders, normalised to the tree amplitude, depend on $\mu_R$ as in Eq. (B1) and agree with the first line of Eq. (23) to $5 \times 10^{-56}$; the released

TABLE VIII. Evaluation time for the two complete Type II coefficients at the benchmark parameters, $c = 0$ and 80-digit working precision, for the unexpanded remainders (exact) and the series. First calls start with empty caches; repeated times are medians of seven evaluations at the same point. Starting the kernel and loading the files are excluded.

| | first call [s] | | | repeated [s] | | |
|---|---|---|---|---|---|---|
| $\beta_A$ | exact | series | ratio | exact | series | ratio |
| 1/10 | 56.04 | 11.61 | 4.83 | 0.3422 | 0.0178 | 19.18 |
| 1/5 | 45.68 | 11.64 | 3.92 | 0.3403 | 0.0179 | 18.99 |
| 3/10 | 44.64 | 11.64 | 3.84 | 0.3485 | 0.0181 | 19.22 |

one-loop files coincide with them at $\mu_R = 1$ GeV. An earlier version of the two-loop renormalisation step, without the factor 64 and with the $O_J$ contribution switched off, was evaluated at $x = 1/11$, $z = 3/11$. At that point, the released $P_1$ remainder equals 64 times this result, multiplied by the tree amplitude, plus the $O_G$–$O_J$ counterterm of the released version. The real parts agree to $5\times10^{-31}$, and the imaginary parts differ by $32C_F$ times the one-loop $O_G$–$O_J$ amplitude, which is a multiple of the tree amplitude. The earlier run is not, however, an alternative result. Its renormalisation constant exceeds $Z_{GG}^2 = Z_{a_s}^2$ by $2\beta_1 a_s^2/\epsilon$, a pole with $C_A^2$ terms that no $O_J$ contribution can supply, and it contains no $O_G$–$O_J$ counterterm. With the same infrared subtraction as the released version, its single poles cannot cancel, so the difference is not an uncertainty of the released remainder. In the released version, the normalisation of the $O_G$–$O_J$ counterterm was fixed by the cancellation of the $C_F n_f$ single pole; a record of the original calculation from June 2019 shows that the same factor is required in both projections at seven kinematic points. The files and a script that reproduce these comparisons are included in the material released with this article.

### Appendix C: Flavour decomposition

To compare the sizes of the flavour contributions, I evaluate the decomposition of Eq. (22) at $n_f = 5$:

$$\begin{aligned}\mathcal{M}_i^{(2)}(5) &= \mathcal{M}_{i,0}^{(2)} + 5\,\mathcal{M}_{i,1}^{(2)} + 5^2\,\mathcal{M}_{i,2}^{(2)} \\ &\equiv a_0 + a_1 + a_2, \qquad i = 1, 2. \end{aligned} \tag{C1}$$

The index $i$ is suppressed in $a_r$. The factors 5 and $5^2$ are included in $a_1$ and $a_2$, respectively, so these quantities are the three terms entering the complete remainder, rather than the unweighted flavour coefficients.

Figure 6 shows the magnitude of each term divided by the magnitude of the complete remainder, $|a_r|/|\mathcal{M}_i^{(2)}(5)|$, at $c = 1/3$, $m_A = \mu_R = 125\,\text{GeV}$ and $N_c = 3$. Panels (a) and (b) correspond to $\mathcal{M}_1$ and $\mathcal{M}_2$, respectively. Since the terms are complex, their magnitudes do not add to the magnitude of their sum. The plotted ratios therefore need not add to unity. A value above the horizontal line at unity means that an individual term is larger in magnitude than the complete remainder and is partly cancelled by the other terms.

The term independent of $n_f$ is the largest contribution to both coefficients over the plotted range. The term linear in $n_f$ is also sizeable. For $\mathcal{M}_1$, it partly cancels the $n_f^0$ contribution. The overall cancellation among the three terms can be quantified by

$$1 - \frac{|a_0 + a_1 + a_2|}{|a_0| + |a_1| + |a_2|}.$$

This quantity is zero when the terms add with the same phase and increases as their sum is reduced by cancellation. For $\mathcal{M}_1$, it lies between 42.2% and 46.9% over the velocities shown. These percentages refer to the reduction in magnitude relative to $|a_0| + |a_1| + |a_2|$, not to a reduction in the cross section.

The relative signs of the flavour-dependent and gluonic terms can also be seen in the leading power coefficients. The combinations $\beta_0 = (11N_c - 2n_f)/3$, $N_c a_A - n_f a_F$ and $4N_c - n_f$ enter $\widehat{A}_{12}$, $\widehat{A}_{11}$ and $\widehat{A}_{21}$, respectively [Eqs. (39) and (40)]. They exhibit the relative minus signs in the logarithmic terms; the size of the cancellation in the complete remainder also depends on their phases and on the powers of $\mathcal{L}$ that they multiply.

Finally, the $n_f^2$ contribution to $\mathcal{M}_2$ starts at $\beta_A^2(1 - c^2)$, as shown in Eq. (35). This suppression explains why its relative contribution in panel (b) becomes small near threshold.

### Appendix D: Evaluation time and released data

The series through $\beta_A^6$ occupy 40 kB, against 204 MB for the published two-loop files, a factor of 5061. The evaluation time was measured before the reduction of Sec. III B, with the unexpanded expressions and the series in the forms used for the comparison of Sec. III C, at $\beta_A = 1/10$, $1/5$ and $3/10$, $c = 0$, $m_A = \mu_R = 125\,\text{GeV}$ and $N_c = 3$, with 80-digit working precision. On a first evaluation the series is 3.84–4.83 times faster. For repeated evaluations at the same point it is 18.99–19.22 times faster, with median 19.18; this does not correspond to a scan over new points. The two timed representations occupy 0.61 MiB and 42.77 MiB on disk, a factor of 70.4. Table VIII gives the individual measurements. The measurements used Wolfram Language 13.2.0 for Linux x86 under Ubuntu 24.04.2 LTS on an AMD Ryzen 9 7950X processor with 124.9 GiB of memory.

The Supplemental Material [28], which is also attached to the arXiv version of this article as ancillary files, is a self-contained package. It contains the construction files, one for each flavour coefficient and power of $\beta_A$ from $\beta_A^{-4}$ to $\beta_A^6$; the series through $\beta_A^6$ in Wolfram Language form; the reductions of the endpoint constants; the 1080 validation records and the forward angle data; the tree-level projections; and an evaluator in Python and Wolfram Language. It also contains the Wolfram Language, C++ and Python code of the threshold construction and of the validation, the outputs of the original runs, against which the construction files and the validation records are checked, and a script that repeats the construction from the published expressions. The published expressions of Ref. [7] are downloaded and checked against checksums by a script. Each table, figure and number in this paper is generated from these data by a single command, including the check of the DPS weight in Fig. 4, the logarithms of Sec. V and the normalisation checks of App. B. A test suite checks the evaluator against the

validation records. Only the construction of the series from the unexpanded expressions requires Mathematica. The series are intended for $\beta_A \leq 0.2$ at the angles tested (Sec. V C); outside this range the unexpanded remainders should be used.


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